A Lightweight Wavelet-Aligned Difference and Mask-Guided Fusion Network for Change Detection

Yuting Liu, *Student Member, IEEE*, Shihua Li, *Senior Member, IEEE*, and Lorenzo Bruzzone, *Fellow, IEEE*

***Abstract*—Remote sensing (RS) change detection (CD) plays an important role in many application domains. To meet the practical demand for both high accuracy and efficiency, and address the fundamental challenges of bitemporal misalignment, insufficient difference representation, and change and nonchange coupling, this article proposes a lightweight wavelet-aligned difference and mask-guided feature fusion network (WDMF-Net). WDMF-Net first employs a Siamese MobileNetV2 with an improved adjacent-level aggregation module (AAM) to extract multiscale complementary features. A wavelet-enhanced feature alignment module (WFAM) is then introduced to enhances local boundaries and global structures of these features in frequency domain, followed by context-consistent spatial alignment to reduce bitemporal misregistration. On this basis, a collaborative feature difference module (CFDM) is designed to model the intermediate complementary relationship between channel and spatial dimensions, enabling accurate localization and capture of contextual differences. Finally, a mask-guided feature enhancement module (MFEM) is employed, where foreground and background mask priors are progressively involved in multiscale difference fusion to explicitly decouple changed and unchanged regions. Experiments demonstrate that WDMF-Net outperforms 16 SOTA methods on five public CD datasets and achieve a favorable balance among detection accuracy, computational efficiency, and generalization capability. The code is available at https://github.com/LYT-Works/WDMF-Net.**

***Index Terms*—Change detection (CD), collaborative feature difference, lightweight, mask-guided feature enhancement, wavelet-enhanced feature alignment.**

I. Introduction

REMOTE sensing (RS) change detection (CD) aims to identify land surface changes from multitemporal RS imagery and plays a crucial role in applications such as urban monitoring [1], resource management [2], and damage assessment [3]. While the increasing availability of high-resolution imagery provides more detailed scene information, it also raises complex background interference and computational costs, thereby placing stringent demands on the accuracy and efficiency of CD methods.

Traditional RS CD methods can be broadly categorized into pixel-based and object-based approaches. Pixel-based methods detect changes using techniques such as image differencing [4], ratio operations [5], principal component analysis [6], and change vector analysis [7]. Despite their simplicity and efficiency, these methods ignore spatial relationships and are sensitive to environmental noise, illumination variations, and radiometric inconsistencies. Object-based methods [2], [8], [9] operate on segmented image objects and detect changes by comparing object-level features, which partially incorporates spatial context. However, their performance heavily depends on segmentation quality and scale selection, often leading to error propagation. Consequently, traditional methods exhibit limited robustness and generalization capability in high-resolution and complex scenes.

In recent years, deep learning (DL) methods have become the dominant paradigm for RS CD. Convolutional neural network (CNN)-based methods [10], [11], [12] demonstrated strong capability in extracting fine-grained differences and delineating change boundaries. Nevertheless, their ability to model long-range dependencies is inherently constrained by the local receptive fields of convolution kernels. To alleviate this limitation, transformer-based [13], [14], [15] and Mamba-based [1], [3], [16] methods have been introduced to enhance contextual difference modeling through self-attention or state space model. Furthermore, hybrid methods [17], [18], [19] that integrate CNN with transformer or Mamba have been explored to jointly capture local details and global context. Despite these advances, achieving effective local–global coordination without introducing feature redundancy or excessive computational overhead remains challenging. Although lightweight CD models [20], [21], [22] have attracted growing practical attention, they often struggle to achieve a favorable trade-off between detection performance and generalization capability.

Based on the above basic network architectures, bitemporal feature extraction strategies can be broadly categorized into single-stream and dual-stream paradigms. Single-stream methods [23], [24] perform early fusion of bitemporal images and learn change representations through unified encoding, which is structurally simple but prone to feature mixing. In contrast,

Received 19 February 2026; revised 16 March 2026; accepted 31 March 2026. Date of publication 3 April 2026; date of current version 16 April 2026. This work was supported in part by the Open Fund of Key Laboratory of Investigation, Monitoring, Protection and Utilization for Cultivated Land Resources, Ministry of Natural Resources, under Grant KLCLR2025KP06; and in part by Sichuan Science and Technology Program under Grant 2026YFHZ0210. *(Corresponding author: Shihua Li.)*

Yuting Liu is with the School of Resources and Environment, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: 202311070410@std.uestc.edu.cn).

Shihua Li is with the School of Resources and Environment, University of Electronic Science and Technology of China, Chengdu 611731, China, and also with the Key Laboratory of Investigation, Monitoring, Protection and Utilization for Cultivated Land Resources, Ministry of Natural Resources, Chengdu 610045, China (e-mail: lishihua@uestc.edu.cn).

Lorenzo Bruzzone is with the Department of Information Engineering and Computer Science, University of Trento, 38123 Trento, Italy (e-mail: lorenzo.bruzzone@unitn.it).

Digital Object Identifier 10.1109/TGRS.2026.3680645

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

dual-stream methods [3], [13], [25] employ a weight-shared Siamese network to extract bitemporal features independently and identify changes through feature interaction, making them widely adopted in CD. However, variations in imaging conditions and terrain relief often cause geometric misalignment between bitemporal images, which hampers the ability of dual-stream methods to capture real changes. To mitigate this issue, existing studies [26], [27], [28], [29], [30], [31] incorporate optical flow estimation or deformable convolutions to enhance feature correspondence. However, most methods primarily focus on spatial-domain local alignment [26], [27], [28] or limited global alignment [29], [30], [31], which is insufficient to jointly model local boundary details and global structural consistency that are essential for precise feature alignment.

Effective difference modeling of aligned features is also critical for CD. Early approaches mainly rely on feature differencing [23], concatenation [24], or distance metrics [32], which are sensitive to noise and background interference. To enhance discriminative representations, subsequent studies introduce attention mechanisms [10], [11], [33] or multibranch designs [18], [34], [35]. Nevertheless, most methods tend to model channelwise or spatialwise differences solely [10], [11], [18], or adopt simple serial [33], [36] or parallel structures with subsequent fusion [34], [35], lacking deep interaction between the two. This limitation hinders accurate localization and identification of difference features. Moreover, multiscale difference fusion during decoding plays a critical role in overall detection performance. Early fusion strategies [13], [23], [25] typically rely on simple upsampling followed by summation or concatenation, which fail to effectively emphasize changed regions across scales. Even when attention mechanisms are incorporated, most existing methods [11], [14], [37], [38] primarily focus on channel or spatial reweighting, without explicitly decoupling changed and unchanged representations, resulting in feature entanglement and pseudo changes.

To address these challenges, we propose a lightweight wavelet-aligned difference and mask-guided feature fusion network (WDMF-Net) that focuses on three fundamental aspects of CD: bitemporal alignment, difference representation, and change and nonchange decoupling. WDMF-Net first extracts multiscale features through hierarchical encoding and aggregation, and achieves contextual bitemporal registration by combining frequency-domain and spatial-domain information. Based on the aligned features, the difference features are collaboratively modeled across both channel and spatial dimensions. Finally, a mask-guided decoder is employed to explicitly disentangle changed and unchanged regions from multiscale difference features. The main contributions are summarized as follows.

1) A lightweight MobileNetV2-based CD framework is proposed. It outperforms many SOTA methods on five public RS CD benchmarks, demonstrating superior performance and strong generalization, and can serve as a robust baseline for lightweight binary CD tasks.

2) A wavelet-enhanced feature alignment module (WFAM) is designed to decompose bitemporal features into multifrequency subbands and performs frequency enhancement prior to spatial alignment. By explicitly decoupling and enhancing local boundary details and global structural components, WFAM enables rigid and context-consistent spatial alignment.

3) A collaborative feature difference module (CFDM) is introduced to model the intermediate complementary relationship across parallel channel and spatial dimensions in the initial difference feature space, enabling accurate localization and capture of local and global differences without introducing excessive attention parameters.

4) A mask-guided feature enhancement module (MFEM) is developed to decouple changed foreground and unchanged background via dual-mask guidance, where masks are not merely final predictions but are progressively exploited as intermediate semantic priors for feature interaction and contextual fusion, thereby enhancing change separability and discriminability.

The rest of this article is organized as follows. Section II reviews the related work. Section III describes the proposed network in detail. Section IV presents the experimental process and results. Section V provides further discussion and analysis. Finally, Section VI concludes the article.

II. Related Work

A. DL-Based CD Methods

CNNs constitute the earliest and most widely explored backbone for RS CD. Caye Daudt et al. [23] established fundamental early-fusion and Siamese fully convolutional paradigms. Peng et al. [24] introduced UNet++ [39] with dense skip connections to enhance the early-fusion features. Fang et al. [25] further integrated channel attention into a Siamese UNet++ to improve change localization. Mainstream studies emphasized bitemporal difference modeling through attention mechanisms and multiscale fusion strategies [10], [11], [30], [34], [38], [40]. Lei et al. [10] enhanced difference features via channel attention and spatial-spectral nonlocal aggregation. Zhang et al. [11] captured implicit differences from concatenated features using coordinate attention and contextual aggregation. Dong et al. [40] embedded normalized Euclidean distance maps into multiscale decoding to strengthen change awareness. In recent years, efficiency-oriented CD methods have attracted increasing attention [20], [41], [42], [43], [44], [45]. Li et al. [20] developed a MobileNetV2 [46]-based CD network that incorporates neighbor feature aggregation and supervised attention. Lei et al. [41] introduced a U-Net [47]-based ultralightweight network that integrates multiscale decoupled convolution and spatial-spectral feature cooperation strategy. Wang et al. [42] proposed an EfficientNet [48]-based network with interleaved feature fusion and bistage decoding, which reduces model parameters while improving detection performance.

Transformers have been extensively introduced to model long-range dependencies in CD. Bandara and Patel [13] proposed ChangeFormer, which encodes bitemporal images using Transformer blocks and predicts change maps via a multilayer perceptron (MLP). Chen et al. [17] combined a ResNet [49] encoder with a transformer decoder for spatio-temporal feature alignment. Yan et al. [14] developed a pure transformer framework, demonstrating the feasibility of

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

fully replacing convolution with self-attention. In addition, foundation models have been introduced into CD to enhance prior knowledge and generalization, such as ChangeCLIP [50] and ChangeViT [19]. To alleviate the high computational cost of self-attention, several lightweight designs have been explored. Shi et al. [51] proposed a hierarchical transformer that combines shallow shifted-window attention with deep deformable attention for efficient computation. Wang et al. [21] introduced a lightweight hybrid network enabling bidirectional local–global feature interaction. Noman et al. [37] employed depthwise separable convolution and pooled-transpose attention for efficient contextual change modeling.

Mambas enable efficient global context modeling with linear complexity and have recently been adopted in CD. Chen et al. [3] first introduced VMamba [52] into CD by combining a Mamba encoder with spatio-temporal decoding. Zhao et al. [16] proposed an omnidirectional selective scan module to alleviate context loss in large-scale RS images. Fu et al. [1] adopted a Siamese VMamba with illumination-aware bitemporal fusion to alleviate low-light feature degradation while reducing model parameters. To further enhance local change details, Zhang et al. [18] incorporated local clues into Mamba-based global modeling to achieve fine-grained binary CD. Zhang et al. [22] adopted a Siamese VMamba with parameter-free interaction and a lightweight CNN module to improve CD performance. Wang et al. [31] proposed a hybrid network that integrates a low-energy DualTime Mamba for global dependency modeling, together with convolutional modules for local edge refinement.

Although Transformer and Mamba offer accurate global modeling capability, their computational overhead remains a concern for high-resolution RS CD. Therefore, this work adopts a lightweight CNN integrated with self-attention to capture local details and global dependencies while satisfying efficiency demands.

B. Flow-Aligned CD Methods

Optical flow was originally developed for adjacent-frame registration in video analysis. Recently, it has been introduced into RS CD to model pixelwise bitemporal displacements, thereby alleviating misregistration caused by viewpoint variations and terrain relief. Fang et al. [26] employed a lightweight FlowNet [53] to achieve bidirectional alignment, followed by feature differencing and fusion to enhance the reliability of changes. Tian et al. [27] estimated displacement offsets from fused bitemporal features and utilized deformable convolution to guide single-temporal feature resampling. Zhang et al. [28] predicted optical flow from concatenated and differenced features for single-direction alignment. As illustrated in Fig. 1, spatial misalignment arises not only at object boundaries but also from global geometric shifts, motivating more context-aware alignment methods for CD. Li et al. [29] enlarged the receptive field for flow estimation and combined it with dynamic deformable convolution to enable more accurate bidirectional alignment. Wang et al. [31] predicted flow fields from wavelet-transformed features to guide the resampling of graph convolutional features for precise structural alignment. Zhou et al. [30] exploited the multiscale feature fusion and skip connections of U-Net to jointly model global and local deformations. Jing et al. [54] predicted the four-vertex offsets of bitemporal images to build a perspective transform, enabling adaptive registration of multiscale features. Unlike existing methods that primarily operate in the spatial domain, the proposed WFAM enhances both local edge details and global structural cues in the frequency domain prior to single-direction spatial alignment, effectively suppressing geometry-induced pseudo changes.

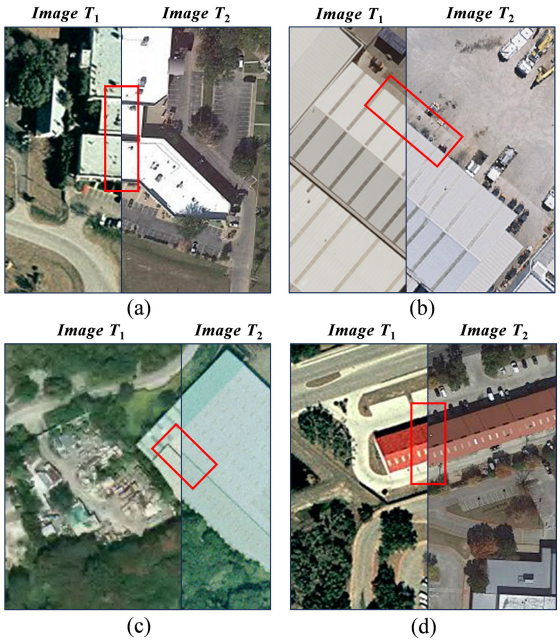


Fig. 1. Examples of geometric misalignment occurring at boundaries and interior regions across different public datasets. (a) LEVIR-CD dataset. (b) WHU-CD dataset. (c) SYSU-CD dataset. (d) LEVIR-CD+ dataset.

C. Mask-Guided CD Methods

Masks were initially introduced in multiclass CD to infer change transitions through bitemporal semantic modeling [3], but have received limited attention in binary CD. Ma et al. [55] proposed a mask-based paradigm for binary CD, in which learnable queries generate adaptive change masks to explicitly constrain context aggregation and emphasize change regions. Xiong et al. [56] combined change masks along with an auxiliary mask loss to refine difference features and suppress pseudo changes. Cui et al. [57] introduced a mask-guided Siamese architecture where change masks guide multiscale feature extraction and fusion to achieve accurate building CD. Ebrahimzadeh and Manzuri [58] adopted a Siamese Mask2Former [59] encoder with a mask-based transformer decoder, where learnable queries are employed to generate change masks for explicit change modeling. Yu et al. [60] integrated cross-level change representation learning with a masked cross-attention transformer, enabling accurate mask generation and change classification. Wang et al. [61] used farmland probability soft mask and binary hard mask generated from bitemporal features to weight and filter difference features, thereby highlighting farmland changes while suppressing nonfarmland interference. Despite their effectiveness,

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

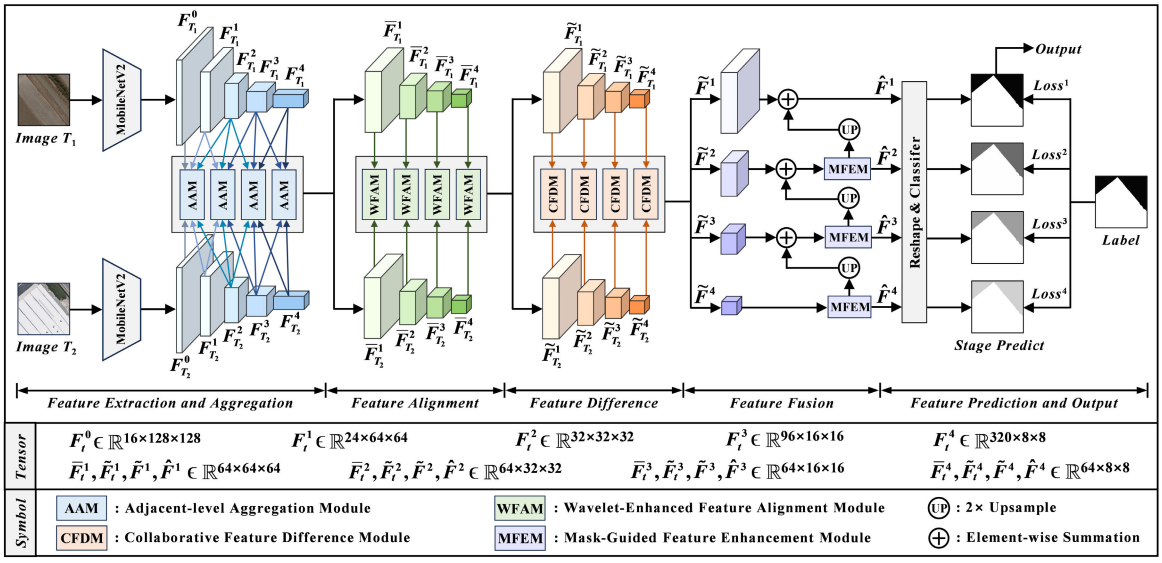


Fig. 2. Overall architecture of the proposed network.

most existing methods rely on a single change mask or task-specific masks for feature modulation, without explicitly and holistically distinguishing changed from unchanged regions. To overcome this limitation, the proposed WFEM generates foreground and background masks directly from the difference features and optimizes contextual fusion under progressively refined dual-mask guidance, thereby enhancing change separability and discriminability.

III. Methodology

A. Overview

As illustrated in Fig. 2, the proposed end-to-end network comprises five sequential stages. First, a Siamese MobileNetV2 is employed as the lightweight backbone to extract multiscale features from the bitemporal RS images. These initial features are further enhanced through an adjacent-level aggregation module (AAM). In the feature alignment stage, WFAM combines frequency-domain enhancement with spatial-domain alignment to strengthen bitemporal frequency representations while alleviating spatial misregistration. During the feature difference stage, CFDM leverages a spatial-channel collaboration mechanism to capture discriminative differences from the aligned features. In the feature fusion stage, MFEM explicitly distinguishes changed and unchanged regions using foreground and background mask-guided attentions. Through progressive upsampling and addition, MFEM effectively integrates deep and shallow difference features. Finally, the fused features are fed into the classification head for multiscale prediction, with deep supervision applied at all hierarchical levels to guarantee accurate binary CD output.

B. Hierarchical Feature Extraction and Aggregation

RS targets exhibit prominent multiscale characteristics: shallow features are detail-rich but noisy, while deep features are semantically informative yet low-resolution. To address this, WDMF-Net employs a Siamese MobileNetV2 with an enhanced AAM for efficient hierarchical feature extraction and aggregation (HFEA).

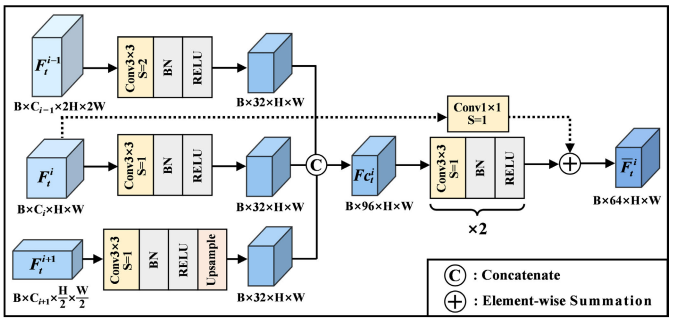


Fig. 3. Architecture of the improved AAM.

MobileNetV2 adopts inverted residual blocks with linear bottlenecks, forming a lightweight and efficient architecture that has been widely used in RS CD tasks [20], [43], [45]. Let the preevent and postevent images be $\boldsymbol{I}_{T_1}$ and $\boldsymbol{I}_{T_2}$. The extracted five-level bitemporal features are denoted as $\boldsymbol{F}_t^i \in \mathbb{R}^{C\times H\times W}(t = T_1, T_2$ and $i = 0, 1, 2, 3, 4)$, with corresponding dimensions illustrated in Fig. 2.

To exploit the multiscale complementary information, an improved AAM is introduced. As depicted in Fig. 3, the low-level feature $\boldsymbol{F}_t^{i-1}$ is downsampled via a 3×3 convolution block with stride 2. Unlike the max pooling (MP) method adopted in [20], [43], and [45], which overlooks subtle yet crucial features by selecting the maximum value, convolutional downsampling employs learnable kernels to better preserve local details. In parallel, the middle-level feature $\boldsymbol{F}_t^i$ is refined by a 3×3 convolution block with stride 1, while the high-level feature $\boldsymbol{F}_t^{i+1}$ is processed by the same convolution block and upsampled to the middle-level resolution. The resulting features from different levels are then concatenated and fused with a residual connection to yield the aggregated output $\bar{\boldsymbol{F}}_t^i$.

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

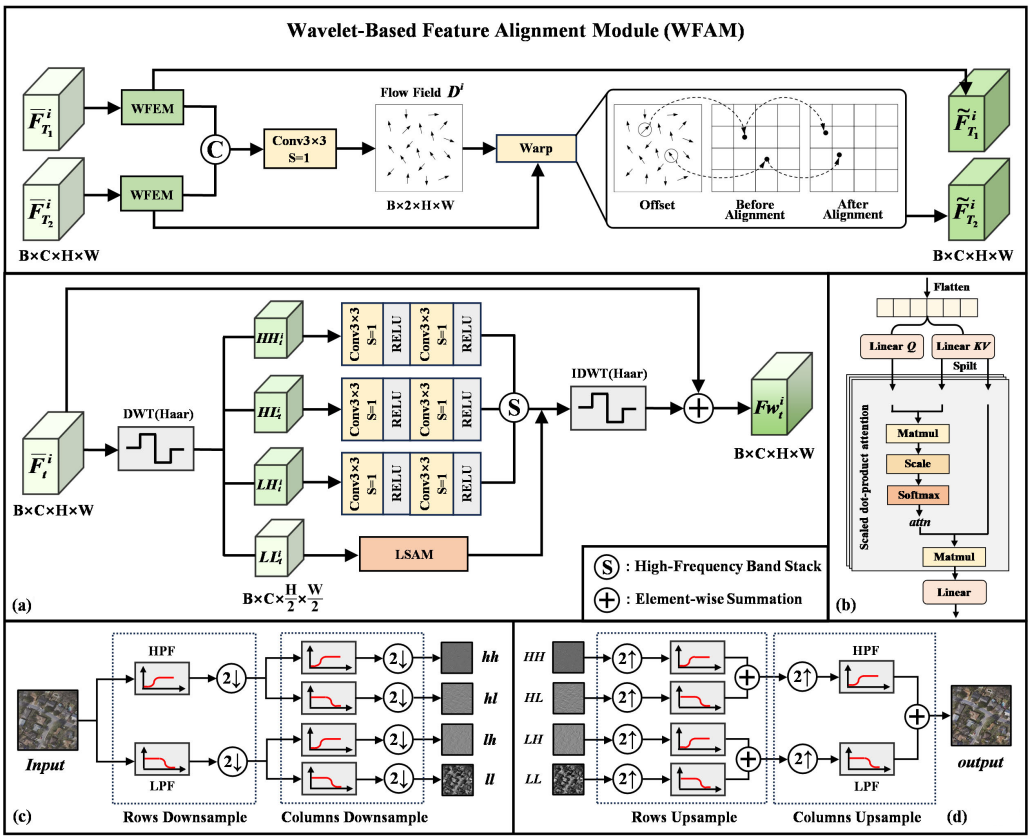


Fig. 4. Architecture of the WFAM. (a) WFEM. (b) LSAM. (c) DWT. (d) IDWT.

Importantly, $\bar{\boldsymbol{F}}_t^1, \bar{\boldsymbol{F}}_t^2$, and $\bar{\boldsymbol{F}}_t^3$ are generated through three-level feature aggregation, whereas $\bar{\boldsymbol{F}}_t^4$ is obtained via a similar two-level aggregation. This design enables the integration of more hierarchical spatial details and semantic information, which can be expressed as

$$\begin{aligned}
\boldsymbol{Fc}_t^i &= \mathrm{Con}\left[\mathrm{CBR}_3^2\left(\boldsymbol{F}_t^{i-1}\right), \mathrm{CBR}_3^1\left(\boldsymbol{F}_t^i\right)\right. \\
&\qquad \left.\mathrm{Up}\left(\mathrm{CBR}_3^1\left(\boldsymbol{F}_t^{i+1}\right)\right)\right], \quad i = 1, 2, 3 \\
\boldsymbol{Fc}_t^i &= \mathrm{Con}\left[\mathrm{CBR}_3^2\left(\boldsymbol{F}_t^{i-1}\right), \mathrm{CBR}_3^1\left(\boldsymbol{F}_t^i\right)\right], \quad i = 4 \\
\bar{\boldsymbol{F}}_t^i &= \mathrm{CBR}_3^1\left(\mathrm{CBR}_3^1\left(\boldsymbol{Fc}_t^i\right)\right) + \mathrm{Conv}_1^1\left(\boldsymbol{F}_t^i\right), \quad i = 12, 3, 4 \quad (1)
\end{aligned}$$

where $\boldsymbol{Fc}_t^i$ and $\bar{\boldsymbol{F}}_t^i$ denote the channel-concatenated feature and aggregated feature, respectively. CBR_3^1 and CBR_3^2 represent a 3×3 convolution with strides of 1 and 2, each followed by batch normalization and ReLU activation. Conv_1^1 refers to a 1×1 convolution for channel adjustment. $\mathrm{Up}(\cdot)$ is the bilinear upsampling operation and $\mathrm{Con}[\cdot]$ is the channel concatenation.

C. Wavelet-Enhanced Feature Alignment Module

Image alignment is essential for mitigating pseudo changes caused by imaging discrepancies. However, most existing CD methods [26], [28], [29] primarily rely on spatial-domain alignment while neglecting frequency-domain information. Frequency-domain modeling provides noise robustness and enables independent enhancement of global structures and local boundaries. Therefore, we introduce the WFAM to precisely optimize and align bitemporal contextual features through spatial and frequency modeling.

As shown in Fig. 4, WFAM consists of two main components: wavelet feature enhancement and flow-based alignment. WFAM first decomposes the input into the frequency domain using the Haar discrete wavelet transform (DWT). Specifically, high-pass filtering (HPF) and low-pass filtering (LPF) are sequentially applied along the row and column directions with downsampling, yielding four subbands

$$\boldsymbol{hh}_t^i, \boldsymbol{hl}_t^i, \boldsymbol{lh}_t^i, \boldsymbol{ll}_t^i = \mathrm{DWT}\left(\bar{\boldsymbol{F}}_t^i\right) \quad (2)$$

where $\bar{\boldsymbol{F}}_t^i$ denotes the aggregated input feature. $\boldsymbol{ll}_t^i$ is the low-frequency subband capturing the global structures, while $\boldsymbol{hh}_t^i, \boldsymbol{hl}_t^i, \boldsymbol{lh}_t^i$ are high-frequency subbands that capture vertical, horizontal, and diagonal edge details, respectively.

For the high-frequency subbands $\boldsymbol{b}_t^i \in \{\boldsymbol{hh}_t^i, \boldsymbol{hl}_t^i, \boldsymbol{lh}_t^i\}$, two successive convolutions are applied to enhance local edge details, which can be formulated as

$$\boldsymbol{B}_t^i = \mathrm{CR}_3^1\left(\mathrm{CR}_3^1\left(\boldsymbol{b}_t^i\right)\right) \quad (3)$$

where $\boldsymbol{B}_t^i \in \{\boldsymbol{HH}_t^i, \boldsymbol{HL}_t^i, \boldsymbol{LH}_t^i\}$ denotes the enhanced high-frequency subbands, CR_3^1 refers to a 3×3 convolution with stride 1, followed by a ReLU activation.

For the low-frequency subband $\boldsymbol{ll}_t^i$, a lightweight self-attention module (LSAM) is introduced to model global dependencies. This subband is first rearranged and linearly projected to obtain the query $\boldsymbol{Q}$ and a shared key-value pair. The pair is split along the channel dimension to form $\boldsymbol{K}$ and $\boldsymbol{V}$,

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

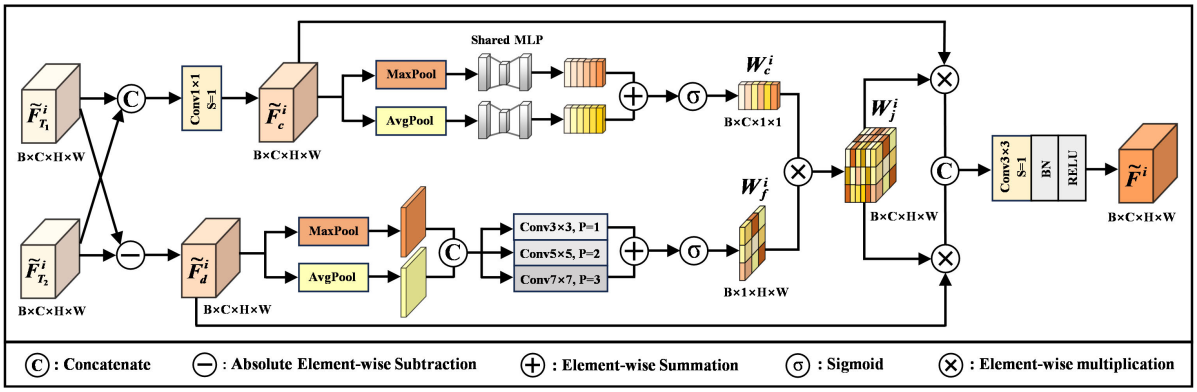


Fig. 5. Architecture of the CFDM.

thereby reducing computational cost while preserving representation consistency. Scaled dot-product attention [62] is then employed to compute the global attention weights, and the output is reshaped back to the original feature space, as formulated in the following equation:

$$\begin{aligned} \boldsymbol{X} &= \text{Rearrange}\left(\boldsymbol{ll}_t^i\right) \\ \boldsymbol{Q} = \boldsymbol{X}\boldsymbol{W}_Q, \boldsymbol{K}, \boldsymbol{Q} &= \text{Spilt}\left(\boldsymbol{X}\boldsymbol{W}_{KV}\right) \\ \boldsymbol{LL}_t^i &= \text{Reshape}\left(\text{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d}}\right)\boldsymbol{V}\right) \end{aligned} \tag{4}$$

where $\boldsymbol{X}$ denotes the reshaped low-frequency subband. $\boldsymbol{W}_Q$ and $\boldsymbol{W}_{KV}$ are learnable projection matrices for generating $\boldsymbol{Q}$, $\boldsymbol{K}$ and $\boldsymbol{V}$. Here, d is the attention-embedding dimension, and the number of attention heads is set to 4. $\boldsymbol{LL}_t^i$ represents the enhanced low-frequency subband.

After subband enhancement, the inverse DWT (IDWT) is applied to progressively upsample and sum four subbands, yielding a reconstructed feature. This feature is added to the WFEM input $\bar{\boldsymbol{F}}_t^i$ to obtain the wavelet-enhanced feature $\boldsymbol{Fw}_t^i$, expressed as

$$\boldsymbol{Fw}_t^i = \text{IDWT}\left(\boldsymbol{HH}_t^i, \boldsymbol{HL}_t^i, \boldsymbol{LH}_t^i, \boldsymbol{LL}_t^i\right) + \bar{\boldsymbol{F}}_t^i. \tag{5}$$

Finally, a flow-based alignment mechanism is introduced in spatial domain to learn bitemporal displacements. Specifically, the wavelet-enhanced features $\boldsymbol{Fw}_{T_1}^i$ and $\boldsymbol{Fw}_{T_2}^i$ are concatenated to predict a dense flow field $\boldsymbol{D}^i \in \mathbb{R}^{2\times \mathrm{H}^i\times \mathrm{W}^i}$, whose two channels represent horizontal and vertical displacements. The predicted flow is then employed in a warping operation to align the postevent feature with the preevent feature coordinates, formulated as

$$\begin{aligned} \boldsymbol{D}^i &= \text{Conv}_3^1\left(\text{Con}\left[\boldsymbol{Fw}_{T_1}^i, Fw_{T_2}^i\right]\right) \\ \tilde{\boldsymbol{F}}_{T_1}^i = \boldsymbol{Fw}_{T_1}^i, \tilde{\boldsymbol{F}}_{T_2}^i &= \text{Warp}\left(\boldsymbol{Fw}_{T_2}^i, D^i\right) \end{aligned} \tag{6}$$

where Conv_3^1 refers to a 3×3 convolution for flow prediction. $\tilde{\boldsymbol{F}}_{T_1}^i$ and $\tilde{\boldsymbol{F}}_{T_2}^i$ are the aligned features, which provide a reliable basis for subsequent difference extraction.

D. Collaborative Feature Difference Module

To address the limitations of single-branch [10], [11], [33], [36] and dual-branch [18], [34], [35] structures that inadequately capture intermediate interactions across channel and spatial differences, the CFDM shown in Fig. 5 introduces a complementary attention mechanism. Specifically, channel attention enhances the channel responses of implicit differences, while spatial attention emphasizes the multiscale spatial distribution of explicit differences, enabling more accurate localization and identification of real changes.

In the channel domain, the aligned features are concatenated and processed by a 1×1 convolution to obtain the channel-fused feature. MP and average pooling (AP) are then applied to capture global spatial statistics. After being mapped by a shared MLP, the pooled features are elementwise summed and sigmoid-normalized to form the channel attention. This process emphasizes difference-sensitive channels while suppressing noise-interfered channels, which can be formulated as

$$\begin{aligned} \tilde{\boldsymbol{F}}_c^i &= \text{Conv}_1^1\left(\text{Con}\left[\tilde{\boldsymbol{F}}_{T_1}^i, \tilde{\boldsymbol{F}}_{T_2}^i\right]\right) \\ \boldsymbol{W}_c^i &= \sigma\left(\text{MLP}\left(\text{AP}\left(\tilde{\boldsymbol{F}}_c^i\right)\right) + \text{MLP}\left(\text{MP}\left(\tilde{\boldsymbol{F}}_c^i\right)\right)\right) \end{aligned} \tag{7}$$

where $\tilde{\boldsymbol{F}}_c^i$ denotes the bitemporal channel-fused feature, $\boldsymbol{W}_c^i$ denotes the channel attention, and $\sigma(\cdot)$ is the sigmoid activation function.

In the spatial domain, an explicit difference feature is first computed by elementwise absolute differencing. MP and AP are then applied and concatenated to form a fused representation, which is processed by multikernel convolutions to accommodate variations in target scales. The resulting features are elementwise summed and normalized to generate the spatial attention. This process jointly captures local and global spatial differences and can be formulated as

$$\begin{aligned} \tilde{\boldsymbol{F}}_d^i &= \left|\tilde{\boldsymbol{F}}_{T_1}^i - \tilde{\boldsymbol{F}}_{T_2}^i\right| \\ \boldsymbol{W}_d^i &= \sigma\left(\sum \text{Conv}_k^1\left(\text{Con}\left[\text{AP}\left(\tilde{\boldsymbol{F}}_d^i\right), \text{MP}\left(\tilde{\boldsymbol{F}}_d^i\right)\right]\right)\right), \quad k = 3, 5, 7 \end{aligned} \tag{8}$$

where $\tilde{\boldsymbol{F}}_d^i$ denotes the bitemporal spatial-difference feature, $\boldsymbol{W}_d^i$ denotes the spatial attention. Conv_k^1 refers to a convolution with $k\times k$ kernel size.

Finally, the two attention maps are multiplied to form a collaborative attention, which is applied to both initial concatenated and difference feature. The weighted features

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

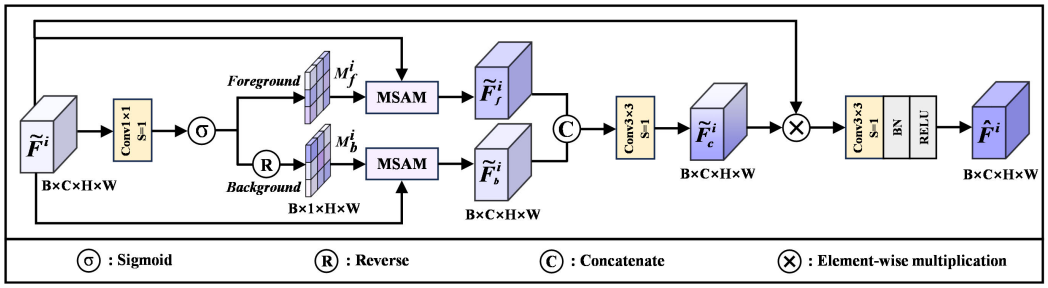


Fig. 6. Architecture of the MFEM.

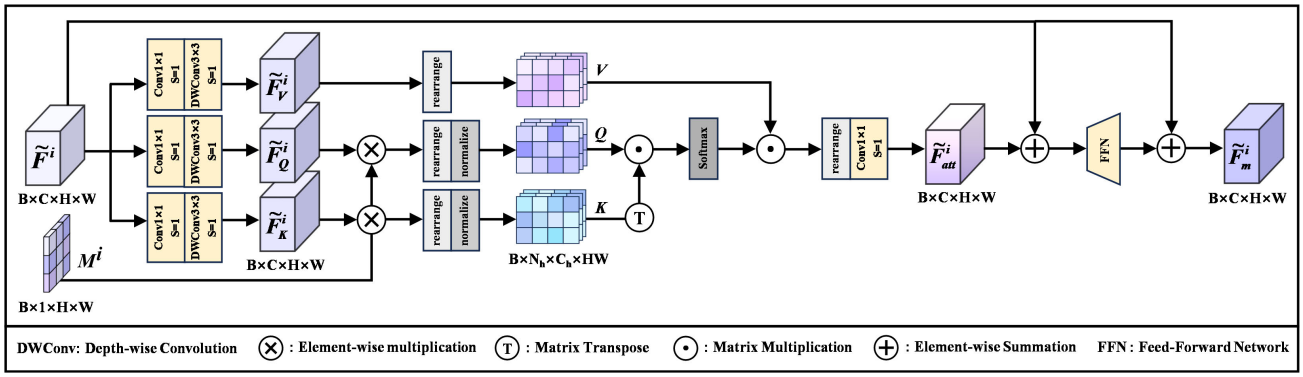


Fig. 7. Architecture of the MSAM.

are then fused by a 3×3 convolution block to produce the discriminative difference feature, expressed as

$$\begin{aligned} \boldsymbol{W}_j^i &= \boldsymbol{W}_c^i \times \boldsymbol{W}_d^i \\ \tilde{\boldsymbol{F}}^i &= \mathrm{CBR}_3^1\left(\mathrm{Con}\left[\boldsymbol{W}_j^i \times \tilde{\boldsymbol{F}}_c^i, \boldsymbol{W}_j^i \times \tilde{\boldsymbol{F}}_d^i\right]\right) \end{aligned} \tag{9}$$

where $\boldsymbol{W}_j^i$ represents the collaborative attention, $\tilde{\boldsymbol{F}}^i$ denotes the difference feature output from CFDM.

E. Mask-Guided Feature Enhancement Module

To mitigate feature mixing between foreground and background regions and address the limitation that most methods [55], [56], [57], [58], [60] rely on a single change mask for feature separation, we introduce the MFEM, as illustrated in Fig. 6. MFEM progressively predicts foreground and background masks as prior constraints and independently leverages self-attention to model global dependencies, thereby improving the separability of changed and unchanged regions.

Given the difference feature $\tilde{\boldsymbol{F}}^i$, MFEM first employs a 1×1 convolution to compress the channel information, followed by a sigmoid activation to generate the foreground mask $\boldsymbol{M}_f^i$. Its reversed form is regarded as the background mask $\boldsymbol{M}_b^i$, which can be defined as

$$\boldsymbol{M}_f^i = \sigma\left(\mathrm{Conv}_1^1\left(\tilde{\boldsymbol{F}}^i\right)\right), \boldsymbol{M}_b^i = 1 - \boldsymbol{M}_f^i. \tag{10}$$

Subsequently, the foreground and background masks guide two parallel mask-guided self-attention module (MSAM) branches to model the global changed and unchanged features, respectively. As shown in Fig. 7, MSAM first employs a 1×1 convolution followed by a 3×3 depthwise separable convolution to encode channel semantics and local spatial contexts into the prior query, key, and value features $\tilde{\boldsymbol{F}}_Q^i, \tilde{\boldsymbol{F}}_K^i, \tilde{\boldsymbol{F}}_V^i$, which can be express as

$$\left\{\tilde{\boldsymbol{F}}_Q^i, \tilde{\boldsymbol{F}}_K^i, \tilde{\boldsymbol{F}}_V^i\right\} = \mathrm{DWConv}_3^1\left(\mathrm{Conv}_1^1\left(\tilde{\boldsymbol{F}}^i\right)\right) \tag{11}$$

where DWConv_3^1 refers to the 3×3 depthwise separable convolution with stride 1.

The corresponding mask is then elementwise multiplied with the prior query and key features to emphasize mask-attended regions. The masked features are reshaped into multihead representations and fed into scaled dot-product attention to capture global dependencies, formulated as

$$\begin{aligned} \boldsymbol{Q}' &= \mathrm{Norm}\left(\mathrm{Rearrange}\left(\tilde{\boldsymbol{F}}_Q^i \times \boldsymbol{M}^i\right)\right) \\ \boldsymbol{K}' &= \mathrm{Norm}\left(\mathrm{Rearrange}\left(\tilde{\boldsymbol{F}}_K^i \times \boldsymbol{M}^i\right)\right) \\ \boldsymbol{V}' &= \mathrm{Rearrange}\left(\tilde{\boldsymbol{F}}_V^i\right) \\ \tilde{\boldsymbol{F}}_{att}^i &= \mathrm{Reshape}\left(\mathrm{Softmax}\left(\frac{\boldsymbol{Q}'\boldsymbol{K}'^{\mathrm{T}}}{\sqrt{d}}\right)\boldsymbol{V}'\right) \end{aligned} \tag{12}$$

where $\tilde{\boldsymbol{F}}_{att}^i$ is the global attention-enhanced feature, and $\boldsymbol{Q}', \boldsymbol{K}', \boldsymbol{V}'$ are the query, key, value matrices guided by the mask $\boldsymbol{M}^i \in \{\boldsymbol{M}_f^i, \boldsymbol{M}_b^i\}$.

The attention-enhanced feature is further refined through residual connections and a feed-forward network (FFN) to enhance nonlinear representation, yielding the MSAM output

$$\tilde{\boldsymbol{F}}_m^i = \mathrm{FFN}\left(\tilde{\boldsymbol{F}}_{att}^i + \tilde{\boldsymbol{F}}^i\right) + \tilde{\boldsymbol{F}}_{att}^i \tag{13}$$

where $\tilde{\boldsymbol{F}}^i$ and $\tilde{\boldsymbol{F}}_m^i$ represent the input and output features of MSAM, respectively.

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

After obtaining the foreground-enhanced and background-enhanced features, they are convolutionally fused to generate a joint feature. This feature is then elementwise multiplied with the original difference feature to perform explicit dual-mask weighting. Finally, a convolutional block is applied to obtain a refined output, expressed as

$$\hat{\boldsymbol{F}}^i = \mathrm{CBR}_3^1\left(\tilde{\boldsymbol{F}}^i \times \mathrm{Conv}_3^1\left(\mathrm{Con}\left[\tilde{\boldsymbol{F}}_f^i, \tilde{\boldsymbol{F}}_b^i\right]\right)\right) \tag{14}$$

where $\tilde{\boldsymbol{F}}_f^i$ and $\tilde{\boldsymbol{F}}_b^i$ are the foreground-enhanced and background-enhanced features, while $\tilde{\boldsymbol{F}}^i$ and $\hat{\boldsymbol{F}}^i$ are the input and output of MFEM, respectively.

F. Loss Function

During model training, a composite loss function is adopted to supervise the learning process of CD tasks. Let $\boldsymbol{P}^i$ denote the prediction at the ith decoding stage, and $\boldsymbol{G}^i$ denote the corresponding ground-truth map, where $\boldsymbol{P}^i, \boldsymbol{G}^i \in [0, 1]$. The binary cross-entropy (BCE) loss is employed to constrain pixelwise prediction and is defined as

$$L_{BCE}\left(\boldsymbol{P}^i, \boldsymbol{G}^i\right) = \frac{-\sum\left[\boldsymbol{G}^i \log\left(\boldsymbol{P}^i\right) + \left(1-\boldsymbol{G}^i\right)\log\left(1-\boldsymbol{P}^i\right)\right]}{N} \tag{15}$$

where N is the number of pixels.

In contrast, the Dice loss measures the consistency between the predicted and ground-truth maps from the perspective of region overlap, and is formulated as

$$L_{Dice}\left(\boldsymbol{P}^i, \boldsymbol{G}^i\right) = 1 - \frac{2\sum\left(\boldsymbol{P}^i \times \boldsymbol{G}^i\right) + \varepsilon}{\sum \boldsymbol{P}^i + \sum \boldsymbol{G}^i + \varepsilon} \tag{16}$$

where ε is a smoothing term introduced to ensure numerical stability. The final composite loss is defined as

$$L = \sum_{i=1}^{4} \lambda_1 L_{BCE}\left(\boldsymbol{P}^i, \boldsymbol{G}^i\right) + \lambda_2 L_{Dice}\left(\boldsymbol{P}^i, \boldsymbol{G}^i\right) \tag{17}$$

where λ_1 and λ_2 are weighting coefficients that balance the contributions of the two loss terms, both set to 1.

IV. Experiments

A. Datasets and Evaluation Metrics

Five widely used high-resolution optical CD datasets are employed, covering diverse spatial resolutions and scene complexities.

1) LEVIR-CD [32] is a building-oriented dataset collected from Google Earth imagery with a spatial resolution of 0.5 m. It consists of 637 bitemporal image pairs of size 1024 × 1024, which are cropped into 256×256 patches, yielding 7120, 1024, and 2048 samples for training, validation, and testing, respectively.

2) WHU-CD [63] is an urban building dataset with a very high spatial resolution of 0.075 m, acquired from aerial imagery. It contains two large images of size 32 507 × 15 354 pixels, which are cropped into 7434 nonoverlapping 256 × 256 patch pairs, including 5946, 744, and 744 samples for training, validation, and testing, respectively.

3) SYSU-CD [64] is a high-resolution dataset with a spatial resolution of about 0.5 m, covering urban and suburban scenes with buildings, roads, and vegetation. It contains 20 000 bitemporal patch pairs, which are divided into 12 000, 4000, and 4000 samples for training, validation, and testing, respectively.

4) LEVIR-CD+ [32] is an extended version of LEVIR-CD with a spatial resolution of 0.5 m, featuring more complex urban scenes and subtler structural changes. It contains 985 bitemporal image pairs, which are cropped into 10 192 patches for training and 5568 patches for testing, with the test set also serving as the validation set.

5) CLCD-CD [65] is a cropland dataset acquired by the Gaofen-2 satellite, with spatial resolutions ranging from 0.5 to 2 m. It originally consists of 600 bitemporal image pairs of size 512 × 512 pixel, which are further cropped into 256 × 256 patch pairs, yielding 1440, 480, and 480 samples for training, validation, and testing, respectively.

Five commonly adopted evaluation metrics are used to quantitatively assess the CD performance, defined as follows:

$$\begin{aligned}
\mathrm{OA} &= \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \\
\mathrm{IoU} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}} \\
\mathrm{P} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\
\mathrm{R} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\
\mathrm{F1} &= \frac{2\times\mathrm{P}\times\mathrm{R}}{\mathrm{P}+\mathrm{R}}.
\end{aligned} \tag{18}$$

Here, TP, FP, TN, and FN indicate the numbers of true positives, false positives, true negatives, and false negatives, respectively.

B. Experimental Setup

The proposed network is implemented in PyTorch 2.1.0 and trained and evaluated on a single NVIDIA GeForce RTX 4090-D GPU. The Adam optimizer [66] is employed with a momentum of 0.9 and a weight decay of 1×10^{-4}, where the exponential decay coefficients β_1 and β_2 are set to 0.9 and 0.99, respectively. An initial learning rate of 5×10^{-4} is used and decayed using a polynomial schedule based on the current iteration. The batch size is fixed at 32. Training is conducted for 40 000 iterations on all datasets, except for the CLCD-CD dataset, which is trained for 20 000 iterations due to its smaller data volume. During preprocessing, data augmentation techniques including random resizing, flipping, and temporal exchange are applied.

C. Comparative Experiments

To evaluate the effectiveness of the proposed method, we compare it with 16 SOTA CD approaches related to bitemporal alignment, difference modeling, and multiscale fusion, covering diverse model types and complexity levels. These include USSFC-Net [41], A2Net [20], SEIFNet [38], EAFH-Net [43], STRobustNet [44], ISDANet [45], BIT [17], ChangerEx [26], BiFA [28], ELGC-Net [37], MaskCD [60], ChangeRD [54],

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE COMPARISON RESULTS ON THE FIVE CONSIDERED DATASETS. RED AND BLUE HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Models	Types	Years	LEVIR-CD	WHU-CD	SYSU-CD	LEVIR-CD+	CLCD-CD
			OA/IoU/F1/R/P(%)	OA/IoU/F1/R/P(%)	OA/IoU/F1/R/P(%)	OA/IoU/F1/R/P(%)	OA/IoU/F1/R/P(%)
USSFC-Net	*C*	2023	99.03/82.36/90.33/92.01/88.71	99.17/84.14/91.39/93.72/89.16	90.47/66.22/79.68/80.12/79.24	98.33/65.36/79.05/81.03/77.17	94.53/45.37/62.42/63.80/61.10
A2Net		2023	99.13/84.14/91.39/90.59/92.20	99.51/90.31/94.91/93.06/96.83	92.32/71.96/83.69/83.54/83.85	98.72/72.74/84.22/83.86/84.58	96.90/63.83/77.92/73.65/82.73
SEIFNet		2024	99.01/81.89/90.04/88.23/91.93	98.99/80.98/89.49/87.44/91.64	91.62/68.80/81.52/78.39/84.91	98.58/69.85/82.25/80.56/84.00	94.89/47.22/64.15/61.39/67.16
EAFH-Net		2024	99.14/84.21/91.43/90.34/92.55	99.56/91.32/95.46/93.91/97.07	92.09/71.46/83.35/84.02/82.70	98.77/73.43/84.68/83.74/85.65	96.62/62.08/76.60/74.36/78.99
STRobustNet		2025	99.08/83.28/90.88/90.05/91.72	99.17/83.87/91.23/87.73/95.02	90.45/66.54/79.91/80.52/79.30	98.73/73.34/84.62/85.93/83.35	93.71/44.68/61.77/68.27/56.39
ISDANet		2025	99.05/82.96/90.69/90.73/90.65	99.07/83.43/90.97/95.14/87.15	86.98/60.61/75.48/84.95/67.90	98.21/65.41/79.08/82.94/75.58	94.11/45.67/62.71/66.59/59.25
BIT	*T*	2021	99.01/82.00/90.11/88.62/91.65	98.31/71.91/83.66/87.88/79.82	89.24/63.56/77.72/79.60/75.93	98.73/72.31/83.93/81.17/86.88	94.36/45.59/62.63/63.48/61.79
ChangerEx		2023	99.15/84.37/91.52/89.75/93.37	99.43/88.79/94.06/91.76/96.48	92.21/69.88/82.27/76.64/88.80	98.84/74.31/85.26/82.47/88.24	96.64/63.85/77.94/79.66/76.29
BiFA		2024	99.06/82.89/90.65/89.36/91.97	99.42/88.72/94.03/92.09/96.05	92.72/72.27/83.90/80.46/87.65	98.63/70.90/82.97/81.88/84.09	96.47/60.81/75.63/73.59/77.78
ELGC-Net		2024	99.06/82.92/90.66/89.22/92.16	98.87/78.16/87.74/81.84/94.57	90.87/66.79/80.09/77.84/82.47	98.71/72.58/84.11/83.81/84.41	95.18/48.14/64.99/60.15/70.69
MaskCD		2024	99.12/83.90/91.24/89.93/92.60	99.48/89.76/94.60/92.21/97.12	92.17/70.51/82.71/79.44/86.25	98.80/74.15/85.16/84.34/86.00	96.79/63.63/77.77/75.45/80.24
ChangeRD		2025	98.99/81.60/89.87/88.22/91.58	98.88/79.27/88.44/86.54/90.42	89.42/61.15/75.89/70.64/81.99	98.77/72.64/84.15/80.12/88.61	95.00/47.01/63.96/59.65/68.93
RSM-CD	*M*	2024	99.12/83.86/91.22/90.20/92.27	99.09/82.46/90.38/86.55/94.57	90.01/64.99/78.78/78.67/78.89	98.67/71.83/83.61/83.42/83.80	95.89/56.11/71.89/70.65/73.17
CDMamba		2025	99.07/82.94/90.67/89.25/92.14	99.33/86.96/93.02/91.05/95.09	90.66/65.69/79.29/75.86/83.05	98.51/69.39/81.93/83.04/80.85	95.79/52.66/68.99/62.94/76.34
RDSF-Net		2025	99.14/84.24/91.44/90.40/92.52	99.11/82.72/90.55/86.28/95.25	90.18/64.45/78.38/75.50/81.49	98.66/70.87/82.95/80.13/85.98	95.81/54.58/70.62/67.63/73.89
GSSR-Net		2025	99.15/84.41/91.55/90.62/92.49	99.33/86.84/92.96/90.10/96.00	90.04/65.21/78.94/79.13/78.75	98.74/73.12/84.48/83.99/84.97	96.14/56.00/71.80/65.96/78.77
WDMF-Net	*C*	2026	99.15/84.45/91.57/91.08/92.07	99.58/91.63/95.63/93.62/97.74	92.55/72.59/84.12/83.69/84.56	98.79/74.45/85.36/86.51/84.24	96.98/64.31/78.28/73.06/84.29

Note: *C*, *T*, and *M* denote CNN-based, Transformer-based, and Mamba-based models, respectively.

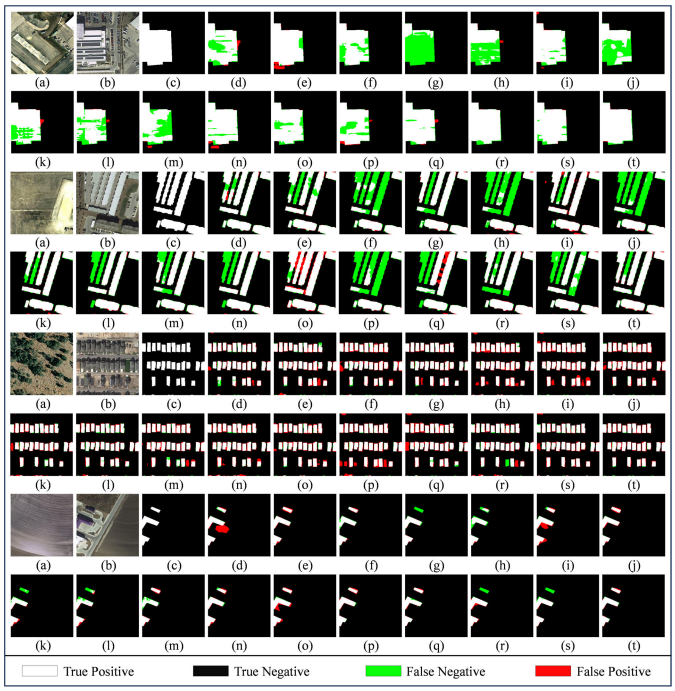


Fig. 8. Qualitative comparison results on the LEVIR-CD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) USSFC-Net. (e) A2Net. (f) SEIFNet. (g) EAFH-Net. (h) STRobustNet. (i) ISDANet. (j) BIT. (k) ChangerEx. (l) BiFA. (m) ELGC-Net. (n) MaskCD. (o) ChangeRD. (p) RSM-CD. (q) CDMamba. (r) RDSF-Net. (s) GSSR-Net. (t) WDMF-Net.

RSM-CD [16], CDMamba [18], RDSF-Net [67], and GSSR-Net [31].

1) Results on the LEVIR-CD Dataset: As shown in Table I, on the LEVIR-CD dataset, WDMF-Net achieves the best performance, with an OA of 99.15%, an IoU of 84.45%, and an $F1$-score of 91.57%, while maintaining a more balanced trade-off between precision and recall.

As illustrated in Fig. 8, in the first industrial scene, most methods provide maps that exhibit missed detections, whereas WDMF-Net accurately captures complete change regions with clear boundaries. In the second strip-shaped building scene, several methods provide maps that exhibit fragmentation and misalignment on slender roofs, while WDMF-Net, benefiting from the WFAM, still produces continuous and precise changes. In the third dense residential area, competing methods are prone to pseudo changes and boundary adhesion due to background interference, whereas WDMF-Net effectively suppresses nonchange responses. In the fourth small-object building sample, although A2Net, SEIFNet, BIT, RSM-CD, and CDMamba can capture subtle changes, WDMF-Net achieves more complete change identification.

2) Results on the WHU-CD Dataset: On the WHU-CD dataset, WDMF-Net achieves an OA of 99.58%, an IoU of 91.63%, an $F1$-score of 95.63%, and a precision of 97.74%, all of which reach the best performance among the compared methods, clearly demonstrating the superiority of the proposed approach.

As shown in Fig. 9, in the first complex building area, most competing methods provide maps that exhibit missed detections and fragmented contours, whereas WDMF-Net better preserves building structures. In the second high-density residential scene with weak change cues, WDMF-Net improves the separability of changed buildings through interactive difference modeling and feature decoupling. In the third strip-shaped road scene, competing methods are prone to missed detections, while WDMF-Net yields continuous and stable results. In the fourth scene with multiscale roofs and illumination variations, WDMF-Net effectively suppresses background noise and highlights real change regions.

3) Results on the SYSU-CD Dataset: On the SYSU-CD dataset, WDMF-Net achieves the best performance with an IoU of 72.59% and an $F1$-score of 84.12%. It maintains a well-balanced trade-off between recall (83.69%) and precision (84.56%), demonstrating strong robustness in complex urban scenes.

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

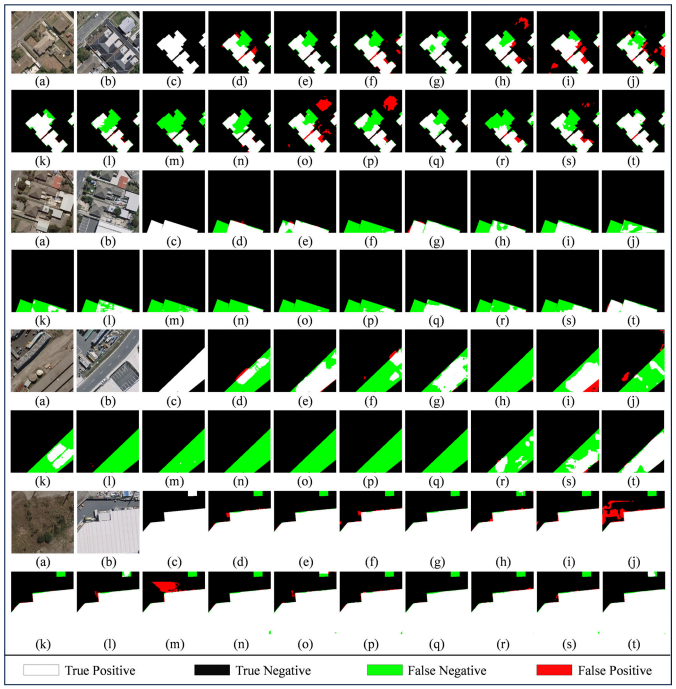


Fig. 9. Qualitative comparison results on the WHU-CD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) USSFC-Net. (e) A2Net. (f) SEIFNet. (g) EAFH-Net. (h) STRobustNet. (i) ISDANet. (j) BIT. (k) ChangerEx. (l) BiFA. (m) ELGC-Net. (n) MaskCD. (o) ChangeRD. (p) RSM-CD. (q) CDMamba. (r) RDSF-Net. (s) GSSR-Net. (t) WDMF-Net.

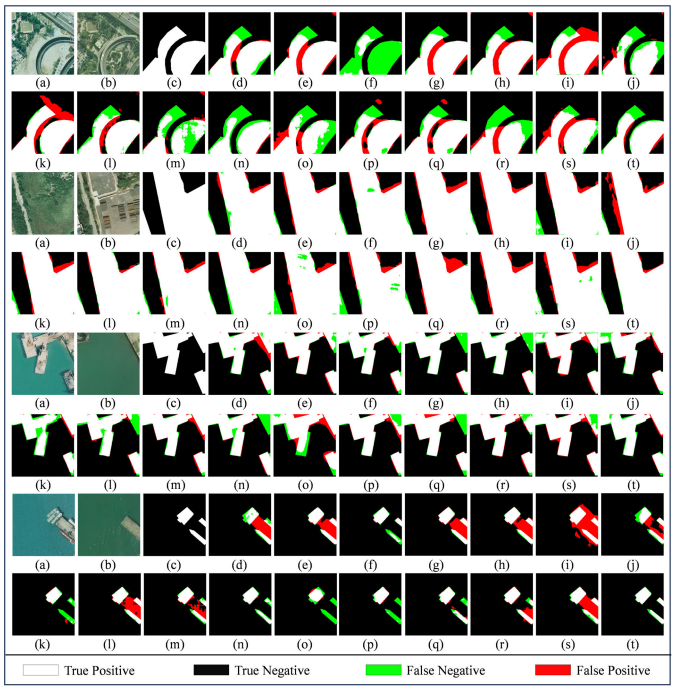


Fig. 10. Qualitative comparison results on the SYSU-CD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) USSFC-Net. (e) A2Net. (f) SEIFNet. (g) EAFH-Net. (h) STRobustNet. (i) ISDANet. (j) BIT. (k) ChangerEx. (l) BiFA. (m) ELGC-Net. (n) MaskCD. (o) ChangeRD. (p) RSM-CD. (q) CDMamba. (r) RDSF-Net. (s) GSSR-Net. (t) WDMF-Net.

As depicted in Fig. 10, in the first curvilinear building area with irregular geometries, WDMF-Net effectively reduces both missed detections and false alarms. In the second bare-land scene, although most competing methods preserve the basic change shapes, WDMF-Net more accurately separates real change regions. In the third harbor area, WDMF-Net reliably delineates the shoreline and preserves the structural integrity of the changed regions. In the fourth small-object ship scene, where changes are easily disturbed by water-surface reflections, other methods tend to produce missed or false detections, whereas WDMF-Net still precisely captures subtle changes.

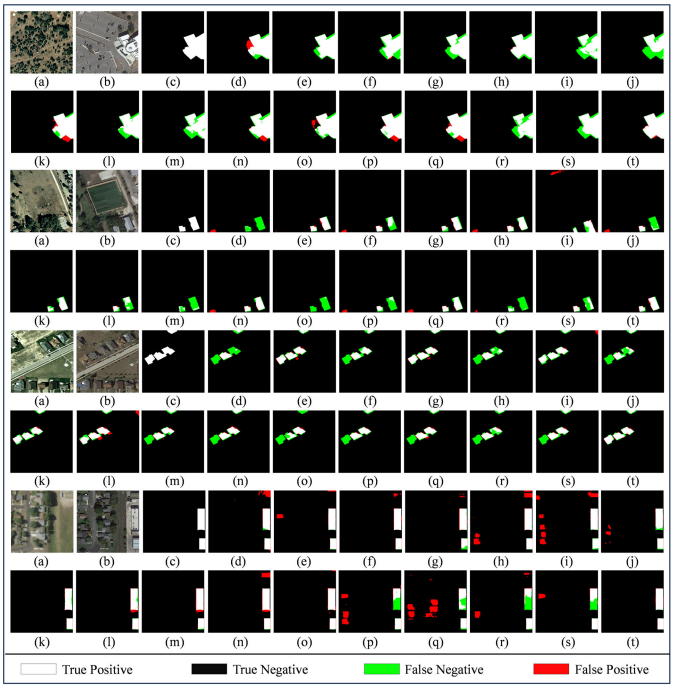


Fig. 11. Qualitative comparison results on the LEVIR-CD+ dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) USSFC-Net. (e) A2Net. (f) SEIFNet. (g) EAFH-Net. (h) STRobustNet. (i) ISDANet. (j) BIT. (k) ChangerEx. (l) BiFA. (m) ELGC-Net. (n) MaskCD. (o) ChangeRD. (p) RSM-CD. (q) CDMamba. (r) RDSF-Net. (s) GSSR-Net. (t) WDMF-Net.

4) Results on the LEVIR-CD+ Dataset: On the LEVIR-CD+ dataset, WDMF-Net achieves the best performance with an IoU of 74.45%, an $F1$-score of 85.36%, and a recall of 86.51%, indicating its ability to detect a larger proportion of real change regions.

As illustrated in Fig. 11, in the first shadow-affected building scene, most competing methods suffer from missed detections and false alarms, whereas WDMF-Net better preserves the integrity of the change regions. In the second small-scale building scene, WDMF-Net effectively separates changed and unchanged areas. In the third regular building scene, although A2Net, EAFH-Net, ISDANet, ChangerEx, and BiFA capture the main changes, WDMF-Net provides more precise contour delineation. In the fourth spectrally inconsistent and low-illumination scene, most methods generate pronounced false alarms, while WDMF-Net more accurately identifies real changes.

5) Results on the CLCD-CD Dataset: On the CLCD-CD dataset, WDMF-Net achieves an OA of 96.98%, an IoU of 64.31%, an $F1$-score of 78.28%, and a precision of 84.29%, ranking first among all compared methods. This confirms its effectiveness in suppressing pseudo changes.

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

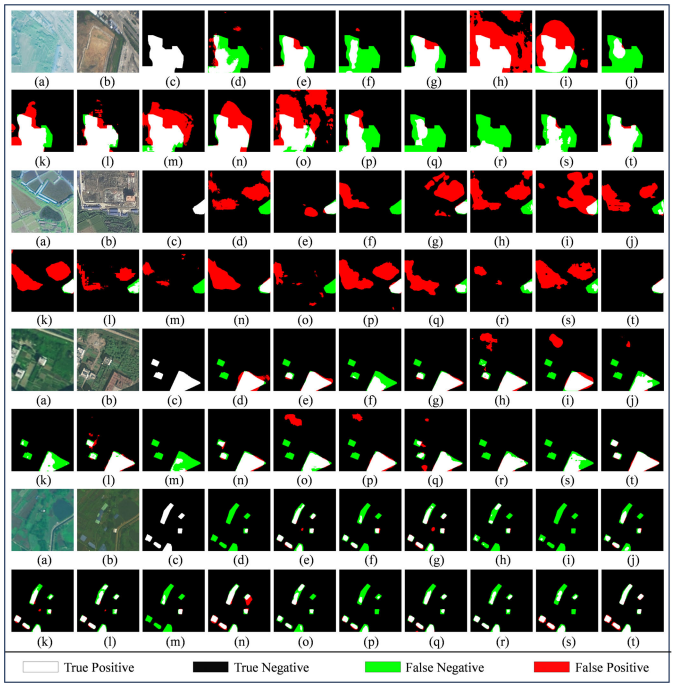


Fig. 12. Qualitative comparison results on the CLCD-CD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) USSFC-Net. (e) A2Net. (f) SEIFNet. (g) EAFH-Net. (h) STRobustNet. (i) ISDANet. (j) BIT. (k) ChangerEx. (l) BiFA. (m) ELGC-Net. (n) MaskCD. (o) ChangeRD. (p) RSM-CD. (q) CDMamba. (r) RDSF-Net. (s) GSSR-Net. (t) WDMF-Net.

As shown in Fig. 12, in the first farmland scene with large illumination differences and irregular field patterns, most methods suffer from evident false alarms and missed detections, whereas WDMF-Net better preserves the integrity of change regions. In the second scenario involving the conversion from farmland to building areas, WDMF-Net effectively identifies changed regions while suppressing background interference. In the third farmland facility scene, despite blurred imagery and weak change cues, WDMF-Net successfully captures subtle changes and maintains structural consistency. In the fourth scene with scattered building changes, most methods exhibit severe missed detections, while WDMF-Net achieves more reliable change representation.

D. Efficiency Analysis

Table II and Fig. 13 report the efficiency comparison in terms of the number of parameters, floating-point operations (FLOPs), and inference time per image pair at an input size of 256 × 256. With only 3.51 M parameters and 4.92 G FLOPs, WDMF-Net exhibits lower computational cost than most transformer-based and Mamba-based methods, while maintaining an inference time of 18.68 ms. Compared with lightweight models that suffer from limited accuracy and high-performance models with excessive computational cost, WDMF-Net achieves the best IoU and $F1$-score across all five datasets, and demonstrates a well-balanced trade-off among accuracy, efficiency, and generalization.

E. Ablation Experiments

The ablation results in Table III demonstrate a progressive performance improvement brought by each module. HFEA yields consistent gains across all datasets, while WFAM and CFDM further improve IoU and $F1$-score. MFEM provides the most significant performance improvement. The qualitative results in Fig. 14 further support these findings. The baseline produces fragmented change regions with noticeable false positives and false negatives. Incorporating HFEA alleviates missed detections, and WFAM further enhances regional completeness. With the integration of CFDM and MFEM, false alarms are effectively suppressed, and the detected change regions progressively approach the ground truth.

To further analyze the feature refinement process, Fig. 15 illustrates the channelwise mean activation maps of intermediate features. When only the baseline is employed, features are mainly concentrated on coarse textures, leading to noise-sensitive representations. After introducing HFEA, bitemporal features become more discriminative and preliminary

TABLE II

EFFICIENCY COMPARISON RESULTS ON THE FIVE CONSIDERED DATASETS. RED AND BLUE HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Models	Params (M)	FLOPs (G)	Inference time(ms)	LEVIR-CD IoU/F1(%)	WHU-CD IoU/F1(%)	SYSU-CD IoU/F1(%)	LEVIR-CD+ IoU/F1(%)	CLCD-CD IoU/F1(%)
USSFC-Net	1.52	4.86	32.63	82.36/90.33	84.14/91.39	66.22/79.68	65.36/79.05	45.37/62.42
A2Net	3.78	3.05	8.42	84.14/91.39	90.31/94.91	71.96/83.69	72.74/84.22	63.83/77.92
SEIFNet	27.91	8.37	5.25	81.89/90.04	80.98/89.49	68.80/81.52	69.85/82.25	47.22/64.15
EAFH-Net	4.81	2.93	10.28	84.21/91.43	91.32/95.46	71.46/83.35	73.43/84.68	62.08/76.60
STRobustNet	5.23	13.29	13.38	83.28/90.88	83.87/91.23	66.54/79.91	73.34/84.62	44.68/61.77
ISDANet	6.40	3.80	28.36	82.96/90.69	83.43/90.97	60.61/75.48	65.41/79.08	45.67/62.71
BIT	3.04	8.75	14.60	82.00/90.11	71.91/83.66	63.56/77.72	72.31/83.93	45.59/62.63
ChangerEx	3.45	8.52	214.64	84.37/91.52	88.79/94.06	69.88/82.27	74.31/85.26	63.85/77.94
BiFA	5.58	53.00	51.15	82.89/90.65	88.72/94.03	72.27/83.90	70.90/82.97	60.81/75.63
ELGC-Net	10.57	187.98	35.37	82.92/90.66	78.16/87.74	66.79/80.09	72.58/84.11	48.14/64.99
MaskCD	107.41	64.09	89.34	83.90/91.24	89.76/94.60	70.51/82.71	74.15/85.16	63.63/77.77
ChangeRD	29.47	18.27	36.12	81.60/89.87	79.27/88.44	61.15/75.89	72.64/84.15	47.01/63.96
RSM-CD	42.30	18.33	37.05	83.86/91.22	82.46/90.38	64.99/78.78	71.83/83.61	56.11/71.89
CDMamba	8.88	54.18	78.76	82.94/90.67	86.96/93.02	65.69/79.29	69.39/81.93	52.66/68.99
RDSF-Net	27.32	15.93	35.13	84.24/91.44	82.72/90.55	64.45/78.38	70.87/82.95	54.58/70.62
GSSR-Net	20.47	13.49	63.88	84.41/91.55	86.84/92.96	65.21/78.94	73.12/84.48	56.00/71.80
WDMF-Net	3.51	4.92	18.68	84.45/91.57	91.63/95.63	72.59/84.12	74.45/85.36	64.31/78.28

TABLE III

QUANTITATIVE ABLATION RESULTS OF KEY MODULES ON THE FIVE CONSIDERED DATASETS

Datasets	Models	HFEA	WFAM	CFDM	MFFM	Params (M)	FLOPs (G)	OA/IoU/F1/R/P (%)
LEVIR-CD	Baseline+					2.29	1.10	98.99/81.48/89.79/87.39/**92.34**
	Baseline+	✓				2.96	2.32	99.03/82.50/90.41/89.83/91.00
	Baseline+	✓	✓			3.20	3.26	99.07/83.13/90.79/89.58/92.03
	Baseline+	✓	✓	✓		3.28	3.71	99.09/83.57/91.05/**91.14**/90.96
	Baseline+	✓	✓	✓	✓	3.51	4.92	**99.15**/**84.45**/**91.57**/91.08/92.07
WHU-CD	Baseline+					2.29	1.10	99.42/88.62/93.97/91.33/96.76
	Baseline+	✓				2.96	2.32	99.47/89.73/94.58/94.01/95.17
	Baseline+	✓	✓			3.20	3.26	99.50/90.28/94.89/93.53/96.29
	Baseline+	✓	✓	✓		3.28	3.71	99.54/91.01/95.30/94.36/96.25
	Baseline+	✓	✓	✓	✓	3.51	4.92	**99.58**/**91.63**/**95.63**/**93.62**/**97.74**
SYSU-CD	Baseline+					2.29	1.10	91.51/69.34/81.89/81.40/82.40
	Baseline+	✓				2.96	2.32	91.79/70.04/82.38/81.36/83.43
	Baseline+	✓	✓			3.20	3.26	92.04/70.60/82.76/80.99/**84.62**
	Baseline+	✓	✓	✓		3.28	3.71	92.10/71.21/83.18/82.87/83.50
	Baseline+	✓	✓	✓	✓	3.51	4.92	**92.55**/**72.59**/**84.12**/**83.69**/84.56
LEVIR-CD+	Baseline+					2.29	1.10	98.53/69.10/81.73/80.97/82.50
	Baseline+	✓				2.96	2.32	98.66/71.29/83.24/81.61/**84.93**
	Baseline+	✓	✓			3.20	3.26	98.70/72.50/84.06/83.80/84.31
	Baseline+	✓	✓	✓		3.28	3.71	98.74/73.14/84.49/84.23/84.74
	Baseline+	✓	✓	✓	✓	3.51	4.92	**98.79**/**74.45**/**85.36**/**86.51**/84.24
CLCD-CD	Baseline+					2.29	1.10	96.15/57.88/73.32/71.01/75.79
	Baseline+	✓				2.96	2.32	96.22/59.19/74.36/73.73/75.00
	Baseline+	✓	✓			3.20	3.26	96.55/60.53/75.41/71.03/80.37
	Baseline+	✓	✓	✓		3.28	3.71	96.56/61.81/76.40/**74.85**/78.01
	Baseline+	✓	✓	✓	✓	3.51	4.92	**96.98**/**64.31**/**78.28**/73.06/**84.29**

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

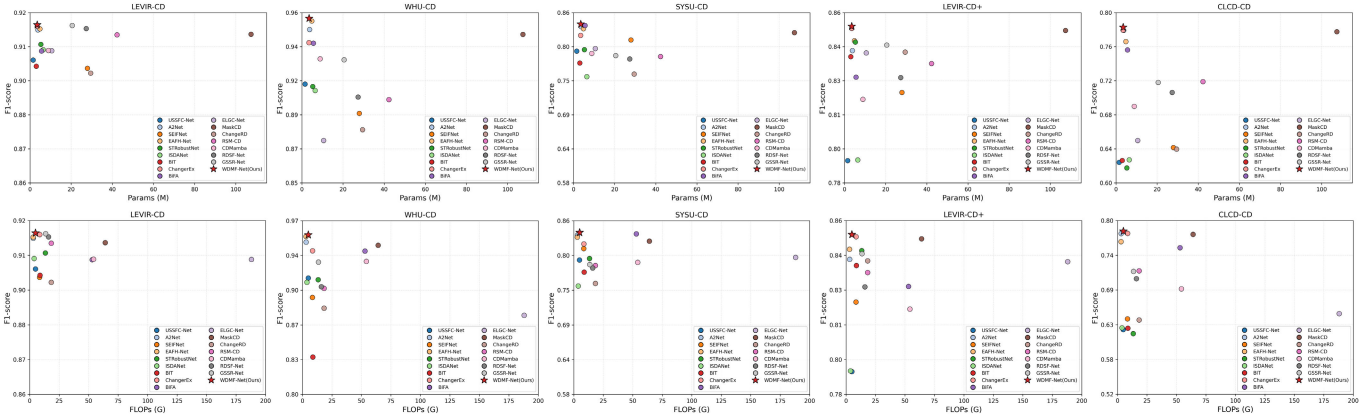


Fig. 13. Performance-efficiency diagrams for all the compared methods across the five considered datasets.

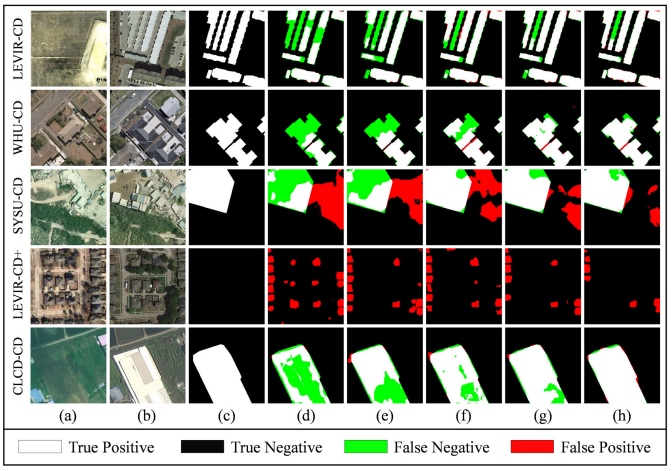


Fig. 14. Qualitative ablation results of the key modules in WDMF-Net on the five considered datasets. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) Baseline. (e) Baseline + HFEA. (f) Baseline + HFEA + WFAM. (g) Baseline + HFEA + WFAM + CFDM. (h) Baseline + HFEA + WFAM + CFDM + MFEM (Ours).

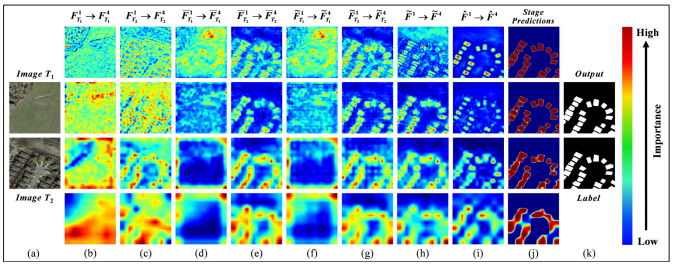


Fig. 15. Four-stage channel-wise mean feature visualizations of progressive module refinement. (a) Bitemporal RGB images. (b) and (c) Bitemporal outputs from MobileNetV2. (d) and (e) Bitemporal outputs from HFEA. (f) and (g) Bitemporal outputs from WFAM. (h) Outputs from CFDM. (i) Outputs from MFEM. (j) Stage predictions. (k) Final output and ground truth.

structural information emerges, though boundary adhesion remains. With the incorporation of WFAM, structural independence is further enhanced. CFDM subsequently produces more compact change regions with clearer and more complete boundaries. Finally, with the integration of MFEM, features are progressively refined to better align with real changes. Overall, predictions at different stages demonstrate improved regional focus, enhanced boundary integrity, and effective noise suppression.

V. Discussion

This section discusses the proposed method from a structural perspective, analyzing the contributions of key components in each module to performance improvements. By comparing different architectural designs across multiple datasets, we highlight the superiority of the proposed design.

A. Discussion on the HFEA

As observed from Table IV and Fig. 16 (#01-#03), replacing the MP followed by convolution strategy in [20], [43], and [45] with convolutional downsampling enables preservation of discriminative structural information. When combined with the five-stage hierarchical feature aggregation mechanism, the proposed model better exploits low-level details than the four-stage design [20], [43], enabling finer characterization of small-scale changes.

Moreover, the comparative results of models with different backbones reported in Table IV (#04-#09) indicate that MobileNetV2 and EfficientNet-B0 achieve superior overall performance, with particularly on the LEVIR-CD dataset. This superiority can be attributed to their well-balanced width-depth configurations, which better support hierarchical feature aggregation. ResNet18 suffers from structural redundancy and inefficient feature utilization, leading to limited performance gains. ShuffleNetV2-1.0× [68] performs competitively on the LEVIR-CD+ dataset with its channel grouping and shuffling strategy, but generalizes less well to other datasets. MiT-B0 emphasizes global modeling and is less sensitive to local changes, limiting its effectiveness within the HFEA module. EfficientVMamba-T [69] performs well on the WHU-CD, SYSU-CD, and LEVIR-CD+ datasets due to its sequential scanning mechanism, but shows weaker adaptability on the LEVIR-CD and CLCD-CD datasets.

B. Discussion on the WFAM

As shown in Table IV and Fig. 16 (#10–#15), removing either wavelet enhancement or feature alignment leads to consistent performance degradation across all datasets, indicating

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE ANALYSIS OF DIFFERENT VARIANTS ACROSS MODULES ON THE FIVE CONSIDERED DATASETS

No.	Models	Params (M)	FLOPs (G)	LEVIR-CD IoU/F1(%)	WHU-CD IoU/F1(%)	SYSU-CD IoU/F1(%)	LEVIR-CD+ IoU/F1(%)	CLCD-CD IoU/F1(%)
(a) HFEA								
#01	w/ Max pooling followed by convolution	3.51	4.92	83.80/91.19	**91.84/95.75**	72.41/83.99	74.30/85.25	61.68/76.30
#02	w/ Four-stage interaction	3.49	4.73	83.95/91.28	91.09/95.34	71.84/83.61	72.25/83.89	62.51/76.93
#03	**Convolutional downsampling with five-stage interaction (Ours)**	3.51	4.92	**84.45/91.57**	91.63/95.63	**72.59/84.12**	**74.45/85.36**	**64.31/78.28**
(b) HFEA with different backbone								
#04	ResNet18	21.76	38.88	82.92/90.66	90.14/94.81	71.77/83.56	73.96/85.03	61.59/76.23
#05	EfficientNet-B0	5.76	6.64	**84.45/91.57**	91.23/95.41	71.62/83.46	74.56/85.43	61.61/76.24
#06	ShuffleNetV2-1.0×	9.54	26.06	83.74/91.15	90.16/94.83	70.76/82.88	**74.93/85.67**	59.87/74.90
#07	MiT-B0	7.55	12.82	83.44/90.97	89.32/94.36	70.75/82.87	73.02/84.41	57.58/73.08
#08	EfficientVMamba-T	11.95	20.17	82.86/90.62	90.87/95.21	71.42/83.33	74.67/85.50	76.70/74.54
#09	**MobileNetV2 (Ours)**	3.51	4.92	**84.45/91.57**	**91.63/95.63**	**72.59/84.12**	74.45/85.36	**64.31/78.28**
(c) WFAM								
#10	w/o Wavelet enhancement	3.26	3.98	83.93/91.27	91.24/95.42	70.53/82.72	72.55/84.09	62.37/76.82
#11	w/o Single-direction alignment	3.51	4.90	83.71/91.13	90.74/95.14	70.92/82.99	72.97/84.37	62.09/76.61
#12	w/ Deformable convolution alignment	3.56	5.02	84.28/91.47	91.29/95.45	70.73/82.86	71.62/83.46	61.47/76.14
#13	w/ Bidirectional alignment	3.60	5.42	84.12/91.38	**92.08/95.88**	72.19/83.85	73.75/84.89	63.47/77.65
#14	Single-direction alignment before wavelet enhancement	3.51	4.92	83.65/91.10	90.98/95.28	71.56/83.42	73.69/84.85	60.77/75.60
#15	**Wavelet enhancement before single-direction alignment (Ours)**	3.51	4.92	**84.45/91.57**	91.63/95.63	**72.59/84.12**	**74.45/85.36**	**64.31/78.28**
(c) CFDM								
#16	Single concatenation branch	3.43	4.51	83.96/91.28	90.74/95.15	68.97/81.64	71.95/83.69	59.22/74.39
#17	Single subtraction branch	3.42	4.47	84.27/91.46	90.43/94.97	70.02/82.36	73.93/85.01	60.68/75.53
#18	Dual branch with addition fusion	3.43	4.51	**84.48/91.59**	91.24/95.42	69.85/82.25	71.01/83.05	61.90/76.46
#19	Dual branch with feature exchange	3.51	4.92	83.60/91.07	91.48/95.55	70.95/83.01	**74.71/85.53**	62.25/76.73
#20	**Dual branch with concatenation fusion (Ours)**	3.51	4.92	84.45/91.57	**91.63/95.63**	**72.59/84.12**	74.45/85.36	**64.31/78.28**
(e) MFEM								
#21	w/o Background mask guidance	3.36	4.11	83.29/90.88	91.51/95.57	71.29/83.24	73.94/85.02	63.46/77.65
#22	w/o Foreground mask guidance	3.36	4.11	83.62/91.08	91.15/95.37	71.29/83.24	73.60/84.80	61.01/75.79
#23	w/o Mask guidance	3.36	4.11	83.88/91.24	91.09/95.34	71.49/83.37	72.91/84.33	61.81/76.40
#24	w/o Self-attention	3.36	3.71	83.84/91.21	91.02/95.30	70.98/83.02	73.41/84.67	61.12/75.87
#25	**Mask guidance with self-attention (Ours)**	3.51	4.92	**84.45/91.57**	**91.63/95.63**	**72.59/84.12**	**74.45/85.36**	**64.31/78.28**
(e) Loss function								
#26	BCE	3.51	4.92	82.45/90.38	89.92/94.69	71.84/83.62	71.10/83.11	62.13/76.64
#27	Dice	3.51	4.92	83.71/91.13	90.57/95.05	70.96/83.01	72.57/84.11	61.79/76.38
#28	0.5 × BCE + 0.5 × Dice	3.51	4.92	82.67/90.51	91.32/95.46	70.94/83.00	72.42/84.01	59.65/74.73
#29	0.5 × BCE + Dice	3.51	4.92	83.91/91.25	90.77/95.16	71.03/83.06	73.32/84.60	60.96/75.75
#30	BCE + 0.5 × Dice	3.51	4.92	83.41/90.96	91.22/95.41	71.16/83.15	73.31/84.60	62.88/77.21
#31	**BCE + Dice (Ours)**	3.51	4.92	**84.45/91.57**	**91.63/95.63**	**72.59/84.12**	**74.45/85.36**	**64.31/78.28**

that relying solely on frequency-domain or spatial-domain is insufficient for accurate change representation. Meanwhile, compared with deformable convolution [27] or bi-directional alignment [26], [29], single-direction alignment effectively alleviates feature misalignment with lower computational cost. When combined with wavelet enhancement, the model achieves more accurate delineation of change boundaries and interior regions. Moreover, performing frequency-domain enhancement provides a more reliable basis for subsequent spatial-domain alignment, enabling more accurate modeling of bitemporal discrepancies.

C. Discussion on the CFDM

As presented in Table IV and Fig. 16 (#16-#20), using a single concatenation or subtraction branch, as illustrated in Fig. 17(a) and (b), provides limited capability for modeling change features. In contrast, the dual-branch structures shown in Fig. 17(c) and (d) introduce complementary information from parallel difference representations. Specifically, simple additive fusion improves detection accuracy and preserves the integrity of change regions to some extent, but still suffers from feature mixing. By comparison, concatenation fusion better preserves discriminative difference cues, enabling the detection of more complete change regions. Notably, the feature exchange strategy, which introduces spatial attention into channel differences and channel attention into spatial differences, may lead to inaccurate feature reweighting, thereby weakening the effectiveness of dual-branch fusion.

D. Discussion on the MFEM

As illustrated in Table IV and Fig. 16 (#21–#25), mask guidance and self-attention mechanisms can improve detection performance, although their effectiveness varies across datasets. Removing either the foreground or background mask leads to consistent performance degradation on all datasets: the absence of background masks tends to introduce false changes, while the absence of foreground masks increases missed detections. Furthermore, when mask guidance is entirely removed, performance on the LEVIR-CD and SYSU-CD datasets does not degrade and even slightly improves. This is mainly because a single mask introduces asymmetric prior constraints, which may overly suppress either foreground or background information. In contrast, the dual-mask guidance provides constraints on changed and unchanged regions, and consistently achieves the best overall performance.

E. Discussion on the Loss Function

The results in Table IV (#26–#31) indicate the best loss configurations for the five considered datasets. On the LEVIR-CD dataset, loss functions incorporating Dice loss

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

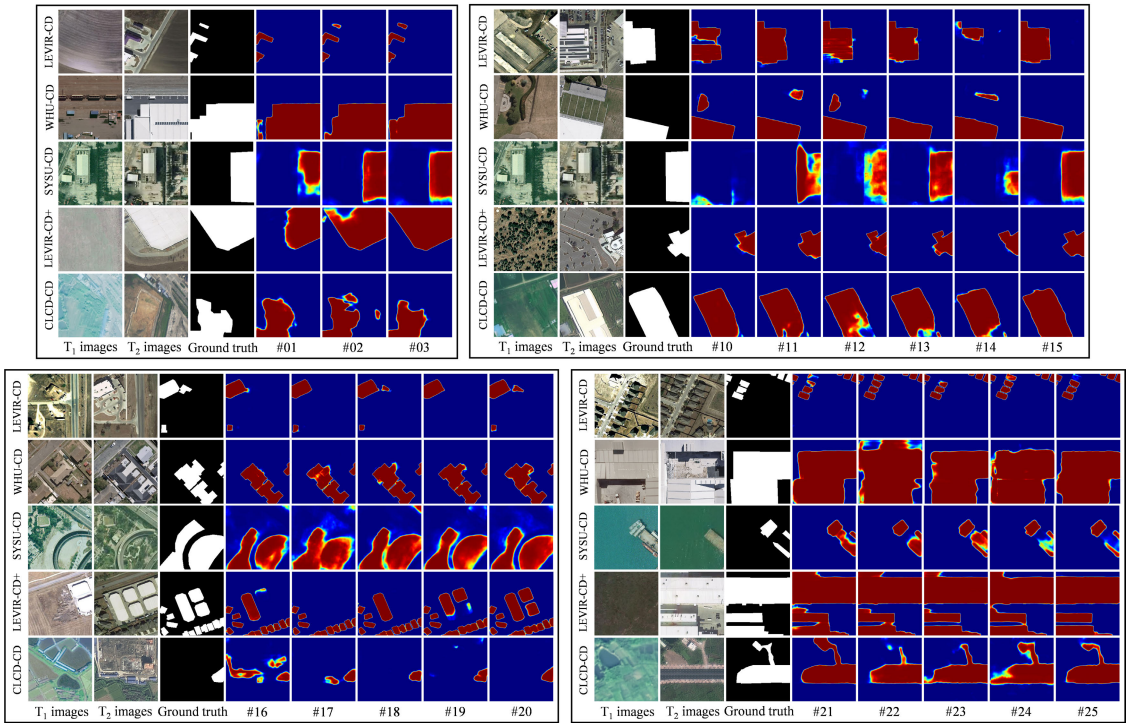


Fig. 16. Qualitative analysis of different variants across modules on the five considered datasets.

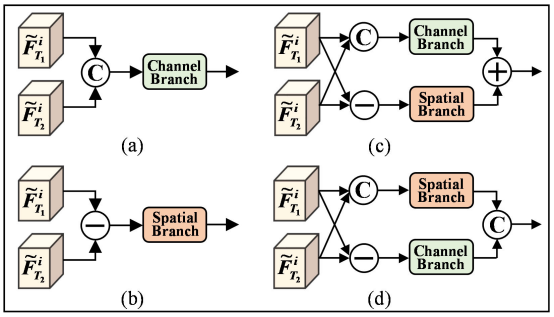


Fig. 17. Simplified schematics of various feature difference modules. (a) Single concatenation branch. (b) Single subtraction branch. (c) Dual branch with addition fusion. (d) Dual branch with feature exchange. Here, $\tilde{F}^i_{T_1}$ and $\tilde{F}^i_{T_2}$ are the aligned features.

constraints consistently outperform BCE loss alone, suggesting that region-level overlap constraint is more critical for this CD scenario. On the WHU-CD and LEVIR-CD+ datasets, various BCE-Dice loss combinations achieve comparable performance, indicating that pixel-level discrimination and region-level consistency contribute with similar importance. For the SYSU-CD dataset, BCE loss alone yields better results than Dice loss and its weighted variants, yet it remains inferior to the combined BCE + Dice loss. On the CLCD-CD dataset, the performance gains brought by BCE loss are more pronounced than those from Dice loss. In summary, the BCE + Dice loss demonstrates stable performance across all datasets.

VI. Conclusion

This article has proposed WDMF-Net, a lightweight CD framework designed to address bitemporal misalignment, insufficient difference representation, and change and nonchange coupling. Within this framework, the modules collaborate to enable feature propagation and complementary enhancement: HFEA first extracts lightweight and robust multiscale features, forming the basis for bitemporal alignment; WFAM performs edge and global frequency enhancement on these features to achieve rigid and context-consistent spatial alignment. Based on the aligned features, CFDM models local and global differences across complementary channel and spatial dimensions, improving change localization and completeness. Finally, MFEM leverages dual mask-guided self-attention to optimize the fusion of multiscale difference features, further improving change separability and discriminability. The four modules complement each other, forming a coherent and efficient framework. Comparative results demonstrate that WDMF-Net achieves superior performance across diverse optical CD scenarios, while maintaining a favorable balance among model size, computational cost, and inference speed. Ablation and discussion analyses further validate the effectiveness and complementarity of the proposed modules.

Despite its satisfactory performance, WDMF-Net relies heavily on supervised annotated data and remains limited in fully capturing complex structural changes. Future work will focus on improving its ability to detect intricate changes and exploring its applicability and scalability in multiclass and multimodal CD tasks.

References

[1] Y. Fu, Z. Wu, Z. Zheng, Q. Zhu, Y. Gu, and M.-P. Kwan, "Mamba-LCD: Robust urban change detection in low-light remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 21200–21212, 2025.

[2] B. Desclée, P. Bogaert, and P. Defourny, "Forest change detection by statistical object-based method," *Remote Sens. Environ.*, vol. 102, nos. 1–2, pp. 1–11, May 2006.

[3] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

[4] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000.

[5] S. H. Winsvold, A. Kääb, and C. Nuth, "Regional glacier mapping using optical satellite data time series," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 8, pp. 3698–3711, Aug. 2016.

[6] M. Dharani and G. Sreenivasulu, "Land use and land cover change detection by using principal component analysis and morphological operations in remote sensing applications," *Int. J. Comput. Appl.*, vol. 43, no. 5, pp. 462–471, Feb. 2019.

[7] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[8] G. Chen, G. J. Hay, L. M. T. Carvalho, and M. A. Wulder, "Object-based change detection," *Int. J. Remote Sens.*, vol. 33, no. 14, pp. 4434–4457, Jan. 2012.

[9] M. Hussain, D. Chen, A. Cheng, H. Wei, and D. Stanley, "Change detection from remotely sensed images: From pixel-based to object-based approaches," *ISPRS J. Photogramm. Remote Sens.*, vol. 80, pp. 91–106, Jun. 2013.

[10] T. Lei et al., "Difference enhancement and spatial–spectral nonlocal network for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4507013.

[11] J. Zhang et al., "AERNet: An attention-guided edge refinement network and a dataset for remote sensing building change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617116.

[12] Y. Liu, S. Li, Z. He, and K. Liu, "Edge and flow guided iterative CNN for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4406422.

[13] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[14] T. Yan, Z. Wan, P. Zhang, G. Cheng, and H. Lu, "TransY-Net: Learning fully transformer networks for change detection of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4410012.

[15] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[16] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-mamba for large remote sensing image dense prediction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5633314.

[17] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[18] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405016.

[19] D. Zhu, X. Huang, H. Huang, Z. Shao, and Q. Cheng, "ChangeViT: Unleashing plain vision transformers for change detection," *Pattern Recognit.*, vol. 172, Apr. 2025, Art. no. 112539.

[20] Z. Li et al., "Lightweight remote sensing change detection with progressive feature aggregation and supervised attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602812.

[21] L. Wang, J. Zhang, and L. Bruzzone, "MixCDNet: A lightweight change detection network mixing features across CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4411915.

[22] M. Zhang, Y. Jiang, and Q. Wang, "Enhancing VMamba for change detection via lightweight feature interaction and selection," *Pattern Recognit.*, vol. 172, Apr. 2026, Art. no. 112648.

[23] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[24] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, Jun. 2019.

[25] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[26] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610111.

[27] J. Tian, D. Peng, H. Guan, and H. Ding, "RACDNet: Resolution- and alignment-aware change detection network for optical remote sensing imagery," *Remote Sens.*, vol. 14, no. 18, p. 4527, Sep. 2022.

[28] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614317.

[29] Y.-C. Li, S. Lei, N. Liu, H.-C. Li, and Q. Du, "IDA-SiamNet: Interactive- and dynamic-aware Siamese network for building change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5628213.

[30] F. Zhou, X. Zhang, H. Shuai, R. Hang, S. Zhu, and T. Geng, "Dual-granularity feature alignment for change detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 4487–4497, 2025.

[31] S. Wang, G. Yuan, and J. Li, "GSSR-Net: Geo-spatial structural refinement network for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5909715.

[32] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[33] M. Yin, Z. Chen, and C. Zhang, "A CNN-transformer network combining CBAM for change detection in high-resolution remote sensing images," *Remote Sens.*, vol. 15, no. 9, p. 2406, May 2023.

[34] A. Eftekhari, F. Samadzadegan, and F. D. Javan, "Building change detection using the parallel spatial-channel attention block and edge-guided deep network," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 117, Mar. 2023, Art. no. 103180.

[35] L. Wang, Y. Fang, Z. Li, C. Wu, M. Xu, and M. Shao, "Summator–subtractor network: Modeling spatial and channel differences for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5604212.

[36] W. Wang, X. Tan, P. Zhang, and X. Wang, "A CBAM based multi-scale transformer fusion approach for remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6817–6825, 2022.

[37] M. Noman, M. Fiaz, H. Cholakkal, S. Khan, and F. S. Khan, "ELGC-Net: Efficient local–global context aggregation for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701611.

[38] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414.

[39] Z. Zhou, M. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.*, vol. 11045, 2018, pp. 3–11.

[40] S. Dong, Y. Zhu, G. Chen, and X. Meng, "EfficientCD: A new strategy for change detection based with bi-temporal layers exchanged," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5639313.

[41] T. Lei et al., "Ultralightweight spatial–spectral feature cooperation network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402114.

[42] M. Wang, B. Zhu, J. Zhang, J. Fan, and Y. Ye, "A lightweight change detection network based on feature interleaved fusion and bistage decoding," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 2557–2569, 2024.

[43] Y. Zhai et al., "Efficient adjacent feature harmonizer network with UAV-CD+ dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5649318.

[44] H. Zhang, Y. Teng, H. Li, and Z. Wang, "STRobustNet: Efficient change detection via spatial–temporal robust representations in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5612215.

[45] H. Ren, M. Xia, L. Weng, H. Lin, J. Huang, and K. Hu, "Interactive and supervised dual-mode attention network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5612818.

[46] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.

[47] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, pp. 234–241.

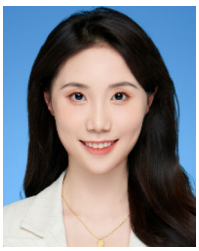
[48] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn. (PMLR)*, vol. 2019, pp. 6105–6114.

[49] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[50] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024.

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.

[51] G. Shi, Y. Mei, X. Wang, and Q. Yang, "DAHT-Net: Deformable attention-guided hierarchical transformer network based on remote sensing image change detection," *IEEE Access*, vol. 11, pp. 103033–103043, 2023.
[52] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.
[53] A. Dosovitskiy et al., "FlowNet: Learning optical flow with convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 2758–2766.
[54] W. Jing, K. Chi, Q. Li, and Q. Wang, "ChangeRD: A registration-integrated change detection framework for unaligned remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 220, pp. 64–74, Feb. 2025.
[55] X. Ma, Z. Wu, R. Lian, W. Zhang, and S. Song, "Rethinking remote sensing change detection with a mask view," 2024, *arXiv:2406.15320*.
[56] F. Xiong, T. Li, J. Chen, J. Zhou, and Y. Qian, "Mask-guided local–global attentive network for change detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3366–3378, 2024.
[57] Y. Cui, H. Chen, S. Dong, G. Wang, and Y. Zhuang, "U-shaped CNN-ViT Siamese network with learnable mask guidance for remote sensing building change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11402–11418, 2024.
[58] M. Ebrahimzadeh and M. T. Manzuri, "MaskChanger: A transformer-based model tailoring change detection with mask classification," in *Proc. 13th Iranian/3rd Int. Mach. Vis. Image Process. Conf. (MVIP)*, Iran, Mar. 2024, pp. 1–6.
[59] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1280–1289.
[60] W. Yu, X. Zhang, S. Das, X. Xiang Zhu, and P. Ghamisi, "MaskCD: A remote sensing change detection network based on mask classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5631316.
[61] H. Wang, N. Wang, and X. Li, "FarmCD: Agricultural information-guided gated network for farmland change detection from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 4401714.
[62] A. Vaswani et al., "Attention is all you need," in *Proc. Conf. Neural Inf. Process. Syst. (NeurIPS)*, Jun. 2017, pp. 6000–6010.
[63] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[64] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.
[65] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.
[66] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[67] S. Wang, D. Cheng, G. Yuan, and J. Li, "RDSF-Net: Residual wavelet mamba-based differential completion and spatio-frequency extraction remote sensing change detection network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 11573–11587, 2025.
[68] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet V2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jul. 2018, pp. 116–131.
[69] X. Pei, T. Huang, and C. Xu, "EfficientVMamba: Atrous selective scan for light weight visual Mamba," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, Apr. 2025, vol. 39, no. 6, pp. 6443–6451.



Yuting Liu (Student Member, IEEE) received the B.S. degree in geographic information science from China University of Mining and Technology, Xuzhou, China, in 2019. She is currently pursuing the Ph.D. degree with the School of Resources and Environment (SRE), University of Electronic Science and Technology of China (UESTC), Chengdu, China.

She is a Visiting Student with the University of Trento, Trento, Italy. Her research interests focus on deep learning for land use and land cover change detection.



Shihua Li (Senior Member, IEEE) received the B.S. and M.S. degrees from Northwest A&F University, Yangling, Xianyang, China, in 1998 and 2001, respectively, and the Ph.D. degree in cartography and geographic information system from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, China, in 2007.

From 2001 to 2004, he was a Lecturer with South China Agricultural University, Guangzhou, China. He is currently a Professor with the University of Electronic Science and Technology of China, Chengdu, China. His research interests include vegetation remote sensing and land use and land cover monitoring.



Lorenzo Bruzzone (Fellow, IEEE) received the Laurea (M.S.) degree (summa cum laude) in electronic engineering and the Ph.D. degree in telecommunications from the University of Genoa, Genoa, Italy, in 1993 and 1998, respectively.

He is currently a Full Professor of telecommunications with the University of Trento, Trento, Italy, where he teaches remote sensing, radar, and digital communications. He is also the Founder and Director of the Remote Sensing Laboratory (RSLab), Department of Information Engineering and Computer Science, University of Trento. He is the Principal Investigator of many research projects and promotes and supervises research activities within numerous national and international programs. Among others, he is the Principal Investigator of the Radar for Icy Moon Exploration (RIME) instrument within the framework of the JUpiter ICy moons Explorer (JUICE) mission of European Space Agency (ESA), Paris, France, and the Science Lead for the High Resolution Land Cover Project within the framework of the Climate Change Initiative of ESA. He is the editor or coeditor of 18 books and conference proceedings and one scientific book. His publications are highly cited, as demonstrated by more than 60 000 citations and an H-index of 117 (source: Google Scholar). He is the author or co-author of more than 390 articles in refereed international journals, more than 390 articles in conference proceedings, and 22 book chapters. His research interests include remote sensing, radar and SAR, signal processing, machine learning, and pattern recognition.

Dr. Bruzzone has been a member of the Administrative Committee of the IEEE Geoscience and Remote Sensing Society (GRSS) since 2009, where since 2019, he has served as the Vice President for Professional Activities and a member of its Permanent Steering Committee. He has been invited as a keynote speaker at more than 40 international conferences and workshops and was a Distinguished Speaker of the IEEE Geoscience and Remote Sensing Society from 2012 to 2016. He is the Co-Founder of the IEEE International Workshop on the Analysis of Multi-Temporal Remote-Sensing Images (MultiTemp) series. He ranked first in the Student Prize Paper Competition at the 1998 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), Seattle, WA, USA. He has received numerous international and national honors and awards, including the IEEE GRSS 2015 Outstanding Service Award, the 2017 and 2018 IEEE IGARSS Symposium Prize Paper Awards, the 2019 WHISPER Outstanding Paper Award, and the 2022 Letter Prize Paper Award for the Best Paper published in IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2022. Since 2003, he has been the Chair of the SPIE Conference on Image and Signal Processing for Remote Sensing. He is the Founder of *IEEE Geoscience and Remote Sensing Magazine*, for which he served as Editor-in-Chief from 2013 to 2017. He has been a Guest Coeditor of numerous special issues of international journals and is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Authorized licensed use limited to: Jiangnan University. Downloaded on October 21,2026 at 12:52:21 UTC from IEEE Xplore. Restrictions apply.